Republic of the Philippines

EASTERN VISAYAS STATE UNIVERSITY

ORMOC CAMPUS

Ormoc City

SOFTWARE REQUIREMENTS SPECIFICATION

BantayGawaMobile: Smart Mobile Workforce Attendance Verification System with Geofencing, Real-Time Facial Recognition, and Workforce Analytics

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1. INTRODUCTION

1.1. Purpose

This Software Requirements Specification (SRS) document specifically describes the BantayGawaMobile system - an intelligent mobile workforce attendance verification system that serves the users from the Environment and Natural Resources Office (ENRO) of Ormoc City City Hall by addressing the problem of time theft and unverified work done among field workers through location based attendance validation using geofencing with real-time facial recognition.

1.2. Scope

BantayGawaMobile is a mobile-first workforce management platform for LGU field workers that only allows them to log attendance if they are physically within the work zones. The system covers:

- User Authentication: Multi-factor authentication with QR Code first-time login, session storage, face recognition.
- Role-Based Access Control: 3 distinct roles with role-specific permissions
- Geofence Management: Circular, Polygon geofences.
- Attendance Management: Time-in, Break-in, Break-out, Time-out with face, geofence validation.
- Field Accomplishment Reporting: Work completion proof submission with geotagged images.
- Leave Request Management: Submit, view, approve/deny leave requests
- Performance Credit System: Star-based worker performance with automated alerts.
- GIS Workforce Map: Real-time worker locator, Work zone status.
- Offline Capability: Attendance, proof capture without internet with auto-sync
- Notification System: In-app messages for announcements, alerts, updates.
- Report Generation: Daily, weekly, monthly report PDF/Excel export.

The system is built using Flutter, Supabase with PostGIS, MobileFaceNet with CNN antispoofing, flutter_map with OpenStreetMap, Firebase Cloud Messaging, hosted on Vercel. System does not integrate with existing LGU payroll & HR as per current scope.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	The overall system administrator with full access to all features
Field Supervisor	An officer assigned to specific work areas who monitors field workers
Field Worker	The frontline worker (e.g., street sweeper) who logs attendance and submits proofs
Geofence	A virtual geographic boundary created using GPS coordinates
Geofencing	Technology that triggers actions when a device enters or exits a defined geographic area
Facial Recognition	Biometric technology that verifies identity by comparing facial features
Liveness Detection	Technology that ensures the face being scanned is from a live person, not a photo/video
Accomplishment Report	Proof of work completed, consisting of an image with embedded metadata
QR Code	Quick Response code used for first-time worker login
GIS	Geographic Information System – used for mapping worker locations and work zones
LGU	Local Government Unit
ENRO	Environment and Natural Resources Office
MFA	Multi-Factor Authentication
JWT	JSON Web Token – used for API authentication
FCM	Firebase Cloud Messaging – used for push notifications
PostGIS	Spatial database extender for PostgreSQL
SQLite	Local database for offline storage on mobile devices
TFLite	TensorFlow Lite – for on-device facial recognition inference

API	Application Programming Interface
REST	Representational State Transfer – API architecture style
CRUD	Create, Read, Update, Delete
SRS	Software Requirements Specification

1.4. Document Overview

This SRS document is organized into the following sections:

1. Introduction
2. Overall Description
3. Use Case Scenarios
4. User Stories
5. Functional Requirements
6. Non-Functional Requirements
7. System Architecture
8. Interface Requirements
9. Data Requirements
10. Testing Strategy
11. Acceptance Criteria

1.5. Sustainable and Development Goals (SDGs) Alignment

This capstone project supports SDG 8: Decent Work and Economic Growth, by ensuring fair and accurate tracking of worker attendance and productivity, preventing time theft and ensuring workers are compensated for actual hours worked. It promotes accountability and transparency in public sector employment. SDG 9: Industry, Innovation, Infrastructure, by making use of innovative technologies (geofencing, facial recognition, GIS mapping) to modernize LGU workforce management. SDG 11: Sustainable Cities and Communities, by ensuring street sweepers and field workers are properly monitored and their work verified. SDG 16: Peace, Justice and Strong Institutions, as the system promotes transparency and accountability in LGU operations, reducing corruption opportunities (time theft, falsified attendance).

1.6. Gender and Development (GAD) Compliance

The system promotes inclusive access for all worker and official personnel regardless of gender, age, ability, or background. The system and its design shall ensure fairness, equality, and usability for everyone.

2. OVERALL DESCRIPTION

2.1. Product Perspective

BantayGawaMobile is a standalone system that operates independently. The system consists of:

1. **Mobile Application (Flutter):** For Field Workers (primary) and Field Supervisors (limited access)
2. **Web Dashboard (Flutter Web):** For Admin and Field Supervisor full management access
3. **Backend Services:** Supabase (authentication, database, real-time) + PostGIS (spatial data)
4. **Facial Recognition Engine:** MobileFaceNet + CNN Anti-Spoofing (on-device, offline)
5. **GIS/Mapping Service:** flutter_map + OpenStreetMap
6. **Notification Service:** Firebase Cloud Messaging (in-app push notifications)
7. **Local Storage:** SQLite (offline data caching and sync)

2.2. Product Features

Feature	Description
Workforce Profile Management	Maintain employee info, assigned work zones, shift schedules, and supervisor assignments
Geofencing-Based Attendance	Employees check in/out only when inside authorized work areas; prevents remote check-ins
Real-Time Facial Recognition	Verifies correct employee is checking in; prevents proxy attendance
Mobile Attendance	Supports Time-in, Break-in, Break-out, Time-out with GPS, timestamp, and face verification
Daily Field Accomplishment Reporting	Employees submit completed tasks with geotagged images (metadata + visual overlay)
Supervisor Validation	Approve attendance, review accomplishments, record observations, add remarks

Workforce Analytics Dashboard	Display attendance rates, late arrivals, absenteeism, field productivity, average working hours
GIS Workforce Map	Show employee locations, assigned work zones, attendance status, completed assignments
Notification Module	Notify employees about work assignments, schedule changes, attendance reminders, supervisor feedback
Executive Dashboard	Display active field workers, attendance trends, productivity reports, delayed reports
Role-Based Access Control	Supports Employee, Supervisor, Admin roles
Offline Mode	Attendance and proof capture without internet; auto-sync when online
QR Code First-Time Login	Workers scan QR code on ID card for seamless first-time authentication
Performance Credit System	Star-based worker performance tracking with automated alerts
Leave Management	Submit, view, approve/deny leave requests with optional proof images
Report Generation	Daily, weekly, monthly reports with PDF/Excel export

2.3. User Classes and Characteristics

User Role	Description	Technical Skill Level	Access Platform	Count (Estimated)
Admin	Top-level system administrator. Manages all users, work areas, geofences, and system settings. Has final authority on all approvals.	High	Web Dashboard + Mobile	1
Field	Field officer assigned to monitor workers	Medium	Web	Approx.

Supervisor	in specific areas. Reviews attendance, proofs, and reports issues to Admin. Can accept/deny proofs for assigned areas only.		Dashboard + Mobile	5-10
Field Worker	Frontline worker (street sweeper) assigned to specific work zones. Logs attendance, submits accomplishment proofs, requests leave.	Low-Medium	Mobile Only	Approx. 50

2.4. Operating Environment

- The system shall run in android 7+ and IOS 10+ capable of using camera, GPS, Internet, SQLite for offline storage.
- The system shall be accessible through any modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari) on desktop or laptop computer.
- The system shall use Supabase for database storage.
- The system shall use Vercel (free tier) for web hosting.
- The system shall use Firebase Cloud Messaging for push notification features,
- The system shall use OpenStreetMap for in-app navigation feature for geofencing.

2.5. Design Constraints

- The system must utilize open-source and free-tier solutions (e.g., Supabase Free Tier and Vercel Free Tier).
- The system must comply with the Data Privacy Act of 2012 (Republic Act No. 10173).
- Core field functions, including attendance recording and proof capture, must remain operational even without an internet connection.
- The mobile application must support Android 7.0 and above. Older devices may have limited face recognition capabilities.
- Facial recognition data must be stored locally on the device and not in cloud storage.
- The system must be deployable using minimal LGU IT infrastructure requirements.
- The system must support English and Filipino, with Cebuano and Waray considered as a future enhancement.

- The geofencing feature must support polygon-shaped boundaries to accommodate irregular work area boundaries.
- All proof images must include embedded date, time, and location information as both image metadata and a visible overlay.

2.6. Assumptions

- Field workers have access to Android 7.0 or later smartphones equipped with a camera and GPS.
- Field workers possess sufficient digital literacy to operate the mobile application.
- Internet connectivity may be intermittent, and offline functionality is sufficient to support field operations during connectivity gaps.
- GPS signals are available in all designated work areas to enable geofence validation.
- The LGU has designated administrative personnel to manage and maintain the system.
- The LGU has assigned Field Supervisors to monitor workers in their designated work areas.
- Workers are willing to use facial recognition and location tracking for attendance verification.
- The LGU provides complete and accurate worker information during the initial system setup.
- The LGU complies with the Data Privacy Act of 2012 (Republic Act No. 10173) and other applicable data protection regulations.
- The Supabase Free Tier is sufficient to support the initial deployment, including approximately 100 workers and 5 supervisors.
- The Vercel Free Tier is sufficient to host the web dashboard during the initial deployment.
- Firebase Cloud Messaging (FCM) services are available and accessible in the Philippines.
- OpenStreetMap map tiles are available for geographic visualization.
- Pre-trained MobileFaceNet and CNN Anti-Spoofing models are available and can be deployed on supported Android devices.
- Payroll integration is not required during the initial deployment of the system.
- Users can understand English and/or Filipino for application navigation.

- Worker devices have cameras capable of capturing facial images with sufficient quality for recognition.
- The LGU can facilitate worker enrollment, including face registration and QR code distribution, during onboarding.

2.7. Dependencies

- The system depends on Supabase for user authentication, database management, and real-time services.
- Spatial data processing depends on the PostGIS extension in PostgreSQL.
- Push notifications depend on the availability of Firebase Cloud Messaging (FCM).
- Map visualization depends on the availability of OpenStreetMap tile servers.
- The web dashboard depends on Vercel hosting services.
- Real-time synchronization and online features depend on internet connectivity.
- Geofence validation depends on the availability and accuracy of GPS signals.
- The mobile application depends on devices running Android 7.0 or later.
- Face recognition functionality depends on the TensorFlow Lite runtime.
- Attendance verification depends on workers completing the face registration process.
- Successful deployment depends on the cooperation and adoption of the system by LGU personnel.
- Initial system setup depends on the successful migration of worker data.
- System deployment depends on LGU approval of privacy policies and related documentation.
- Effective system adoption depends on the successful completion of user training.
- Future use of paid cloud services depends on LGU budget approval.
- Certain system functions depend on the availability and reliability of third-party services and APIs.

3. USE CASE DEVELOPMENT

3.1. UC-01 - Create User Account

Actor: Admin

Goal: To create a new user account for a Field Worker or Field Supervisor and register their facial data for biometric verification.

Precondition: The Admin is logged in with valid credentials and has the necessary permissions to create user accounts.

Main Flow:

1. The Admin navigates to Employee Management page.
2. The Admin selects "Add Employee".
3. The Admin enters employee details (Name, Role, Contact Information).
4. The Admin selects the employee's role (Field Worker or Field Supervisor).
5. The Admin uploads or uses the system camera to capture the employee's face.
6. The system converts the captured face into a face embedding (feature vector) .
7. The system securely stores the face embedding in the database .
8. The system automatically generates a unique QR code for the employee.
9. The Admin assigns the employee to a work area (if applicable).
10. The Admin saves the account.
11. The system creates the account and sends a welcome notification.

Alternative Flow:

1. If the Admin submits incomplete information, the system prevents submission and highlights all required fields.
2. If face capture fails, the system allows the Admin to retry.

Postcondition: The user account is created in the system; the QR code is generated and available for printing; the face embedding is stored securely for future biometric verification.

3.2. UC-02 - Manage Work Areas and Geofences

Actor: Admin

Goal: To create, edit, or delete work areas with defined geofence boundaries (polygon or circle) for attendance restriction

Precondition: The Admin is logged in and has access to the Area Management module

Main Flow:

1. The Admin navigates to Area Management page.
2. The Admin views the interactive map with existing geofences.
3. The Admin selects "Add Area".
4. The Admin draws a geofence on the map (polygon or circle).
5. The Admin sets the area name and other properties.
6. The Admin assigns a Field Supervisor to the area.
7. The Admin assigns Field Workers to the area.
8. The Admin saves the changes.
9. The system creates the geofence and assigns the workers and supervisor.

Alternative Flow:

1. If the Admin draws an invalid geofence (self-intersecting polygon), the system prevents creation and prompts the Admin to redraw.
2. If the Admin attempts to delete an area with active workers assigned, the system warns and requires confirmation before proceeding.
3. If the Admin edits an existing geofence, the system updates the boundary and notifies all assigned workers.

Postcondition: The work area with geofence is created/updated/deleted in the system. The assigned workers can now log attendance only within this boundary.

3.3. UC-03 - Approve or Deny Leave Request

Actor: Field Supervisor

Goal: To review, approve, or deny a leave request submitted by a Field Worker

Precondition: The Admin is logged in. A leave request has been submitted by a Field Worker

Main Flow:

1. The Admin navigates to Leave Management page.
2. The Admin views the list of pending leave requests (filtered by area).
3. The Admin selects a leave request.
4. The Admin reviews the worker's details, reason for leave, and optional proof image.
5. The Admin selects "Approve" or "Deny".
6. If denying, the Admin enters a reason for denial.
7. The system updates the leave request status.
8. The system sends a notification to the worker and their Field Supervisor.

Alternative Flow:

1. If the Admin denies the request without providing a reason, the system prompts the Admin to enter a reason.

Postcondition: The leave request is approved or denied; the worker and Supervisor are notified of the decision

3.4. UC-04 - Review and Approve Accomplishment Proof

Actor: Field Supervisor

Goal: To review, approve, or deny an accomplishment proof submitted by a Field Worker.

Precondition: The user is logged in, an accomplishment proof has been submitted by a Field Worker. For the Field Supervisor, the proof must be from an assigned area.

The user is logged in; an accomplishment proof has been submitted by a Field Worker; for Field Supervisor, the proof must be from an assigned area.

Main Flow:

1. The user navigates to Accomplishment Management page.
2. The user views the list of pending proofs (filtered by area for Supervisor; all areas for Admin).
3. The user selects a proof.
4. The system displays the proof image with visual timestamp and coordinates overlay.
5. The system displays the extracted metadata summary (date, time, location).
6. The system indicates if the metadata matches the assigned area.
7. The user selects "Approve" or "Deny".
8. If denying, the user enters a reason.
9. The system updates the proof status.
10. The system sends a notification to the worker and the other approver (Supervisor or Admin).

Alternative Flow:

1. If the user is Field Supervisor and the field worker who submits is not from their assigned area, the system does not display the accomplishment report.
2. If the metadata extracted from the image does not match the assigned area, the system flags this and displays a warning to the user.
3. If the other approver has already approved the proof, the current user can request an appeal.

Postcondition: The accomplishment proof is approved or denied; the worker and other approver are notified.

3.5. UC-05 - Generate Reports

Actor: Admin or Field Supervisor

Goal: To generate and export attendance or performance reports.

Precondition: The user is logged in.

Main Flow:

1. The user navigates to Reports page.
2. The user selects a report type (Attendance Summary, Area Performance, Worker Performance, Accomplishment Report).
3. The user applies filters (date range, area, worker, status).
4. The system displays a preview of the report.
5. The user selects export format (Excel, PDF, or CSV) 6. The system generates and downloads the report

Alternative Flow:

1. If the user is a Field Supervisor, the system only shows areas assigned to them.
2. If no data matches the selected filters, the system displays a "No data found" message.

Postcondition: The report is generated and downloaded in the selected format.

3.6. UC-06 - Send Notification

Actor: Admin

Goal: To send in-app notifications to users for announcements, warnings, or confidential feedback.

Precondition: The Admin is logged in.

Main Flow:

1. The Admin navigates to Notifications page.
2. The Admin composes a message.
3. The Admin selects recipients (All Users, All Supervisors, All Workers, Specific User).
4. The Admin marks the notification as "Confidential" if needed (only recipient sees).
5. The Admin sends the notification.
6. The system delivers the in-app notification to selected recipients via Firebase Cloud Messaging.

Alternative Flow:

1. If the Admin selects "Confidential" for a notification, the system ensures only the intended recipient can view it.

2. If the Admin sends a notification without a message, the system prevents sending.

Postcondition: The notification is sent to all selected recipients.

3.7. UC-07 - Shuffle Worker Assignments

Actor: Admin

Goal: To reassign Field Workers to different work areas, typically done monthly.

Precondition: The Admin is logged in; the monthly shuffle is due.

Main Flow:

1. The Admin navigates to Area Management page.
2. The Admin views all work areas with current worker assignments.
3. The Admin selects a work area.
4. The Admin views the list of available workers.
5. The Admin selects workers to assign to this area.
6. The Admin confirms the assignment.
7. The system updates the worker assignments.
8. The system sends notifications to affected workers.

Alternative Flow:

1. If the Admin attempts to assign a worker who is already assigned to another area, the system warns and prompts confirmation.
2. If the Admin selects "Bulk Assign," the system allows assigning multiple workers to multiple areas simultaneously.

Postcondition: Worker assignments are updated; affected workers receive notifications of their new assignments.

3.8. UC-08 - View GIS Workforce Map

Actor: Admin or Field Supervisor

Goal: To view real-time worker locations and work zone status on an interactive map.

Precondition: The user is logged in.

Main Flow:

1. The user navigates to GIS Map page.
2. The system displays an interactive map with: - Worker markers showing real-time locations - Color-coded attendance status (On-site/Off-site) - Geofence overlays for all or assigned areas - Area names and boundaries.
3. The user clicks on a worker marker.

4. The system displays worker details (name, area, status, last check-in).

Alternative Flow:

1. If the user is a Field Supervisor, the system only displays areas assigned to them.
2. If the user enables "Historical Movement" toggle, the system displays worker movement paths for the day.

Postcondition: The user has real-time visibility of worker locations and zone status.

3.9. UC-09 - View and Respond to Alerts

Actor: Admin

Goal: To view and respond to system-generated alerts for consecutive absences, credit warnings, and breaches.

Precondition: The Admin is logged in; alerts have been generated by the system.

Main Flow:

1. The Admin navigates to Notifications/Alerts page.
2. The Admin views alerts categorized by type: - Consecutive Absence Alerts (3+ absences) - Credit Score Warnings (Yellow/Red) - Breach Alerts.
3. The Admin selects an alert.
4. The system displays affected worker details.
5. The Admin sends a warning notification to the worker and Supervisor.
6. The system logs the alert response.

Alternative Flow:

1. If the Admin dismisses an alert without action, the system logs the dismissal for audit purposes.

Postcondition: The alert is acknowledged and responded to; the response is logged in the system.

3.10. UC-10 - Monitor Assigned Workers (Field Supervisor)

Actor: Field Supervisor

Goal: To monitor the attendance and location status of Field Workers assigned to the Supervisor's areas

Precondition: The Field Supervisor is logged in.

Main Flow:

1. The Supervisor navigates to the Dashboard.
2. The Supervisor views the list of assigned areas.

3. The Supervisor views the team attendance summary.
4. The Supervisor views the GIS map showing worker locations in assigned areas.
5. The Supervisor clicks on a worker marker.
6. The System displays the worker's profile with attendance history

Alternative Flow:

1. If the Supervisor filters workers by area, the system displays only workers in that area

Postcondition: The Supervisor has full visibility of all assigned workers' attendance and location status.

3.11. UC-11 - Report Worker Issue

Actor: Field Supervisor or Field Worker

Goal: To report an issue about another worker's conduct, absence, or performance to the appropriate authority.

Precondition: The user is logged in, and an issue has been observed.

Main Flow:

1. The user navigates to the Report Worker page.
2. The user selects the worker to report
3. The user selects an issue type (e.g., not working, outside assigned area, poor performance).
4. The user adds a description of the issue.
5. The user optionally attaches a proof image.
6. The user submits the report.
7. The system sends a notification to the Supervisor (if Worker reports) or Admin (if Supervisor reports).

Alternative Flow:

1. If the user submits a report without a description, the system prevents submission and prompts for a description.

Postcondition: The report is sent to the appropriate authority for review.

3.12. UC-12 - Generate Reports (Supervisor)

Actor: Field Supervisor

Goal: To generate and export reports specific to the Supervisor's assigned areas.

Precondition: The Field Supervisor is logged in.

Main Flow:

1. The Supervisor navigates to Reports page.
2. The Supervisor selects a report type (Attendance Summary, Area Performance).
3. The Supervisor applies filters (date range, area, worker).
4. The system displays a preview of the report.
5. The Supervisor selects export format (Excel, PDF, or CSV).
6. The Supervisor optionally sends the report to Admin.
7. The system generates and downloads/sends the report.

Alternative Flow:

1. If the Supervisor attempts to generate a report for an unassigned area, the system does not show that area.

Postcondition: The report is generated and exported or sent to Admin.

3.13. UC-13 - View Leave Requests (Supervisor)

Actor: Field Supervisor

Goal: To view leave requests submitted by workers assigned to the Supervisor's areas.

Precondition: The Supervisor is logged in.

Main Flow:

1. The Supervisor navigates to Leave Management page.
2. The Supervisor views the list of leave requests from assigned workers.
3. The Supervisor selects a request to view details.
4. The system displays the request details (worker name, reason, dates, proof image).
5. The Supervisor notes they cannot accept/deny (Admin only).
6. The Supervisor optionally adds comments for Admin's consideration

Alternative Flow:

1. If the Supervisor filters leave requests by status, the system displays only requests matching the filter.

Postcondition: The Supervisor is informed of all leave requests from assigned workers.

3.14. UC-14 - First-Time Login via QR Code

Actor: Field Worker

Goal: To log in to the system for the first time by scanning the QR code printed on the worker's ID card.

Precondition: The Admin has created the worker's account and face embedding; the QR code is printed on the worker's ID card or available digitally.

Main Flow:

1. The worker downloads the mobile application.
2. The worker opens the app and sees the "Scan ID to Login" button.
3. The worker taps the button.
4. The app opens the camera.
5. The worker scans the QR code on the ID card.
6. The app authenticates the worker.
7. The system stores the session persistently.
8. The worker is logged in and sees the Dashboard

Alternative Flow:

1. If the worker scans an invalid QR code, the system displays an error and prompts the worker to try again.
2. If the worker has saved the QR code as an image, they can choose to scan from the gallery.

Postcondition: The worker is logged in; the session is saved for all future use.

3.15. UC-15 - Log Attendance

Actor: Field Worker

Goal: To log attendance (Time-in, Time-out) using facial recognition and geofence validation.

Precondition: The worker is logged in; the worker is within the assigned geofence boundary.

Main Flow:

1. The worker opens the app (auto-logged in).
2. The worker navigates to the Attendance page.
3. The worker selects an attendance action: - Time-in - Time-out.
4. The system checks if the worker is inside the assigned geofence.
5. The system performs facial recognition (on-device).
6. The system verifies the captured face matches the registered face embedding.
7. If both checks pass, the system logs attendance with: - Timestamp - GPS coordinates - Selfie proof.
8. The system updates the attendance status on the dashboard.

Alternative Flow:

1. If the worker is outside the assigned geofence, the system displays "Outside geofence" and does not allow attendance logging.
2. If facial recognition fails, the system uses the built-in camera to capture a selfie with embedded date, time, and location.
3. If there is no internet connection, the system caches the attendance locally and syncs when connectivity is restored.

Postcondition: Attendance is recorded (or cached for offline sync) in the system.

3.16. UC-16 - Submit Accomplishment Proof

Actor: Field Worker

Goal: To submit proof of completed work via image capture with embedded location, date, and time metadata.

Precondition: The worker is logged in, and work has been completed.

Main Flow:

1. The worker navigates to the Accomplishment page.
2. The worker taps "Upload Today's Report" (default assigned area).
3. The worker opens the app's built-in camera.
4. The worker captures a proof image.
5. The system automatically: - Embeds date, time, and location coordinates in metadata - Burns visual overlay on the image (date, time, coordinates).
6. The worker adds an optional message.
7. The worker submits the proof.
8. The system validates the metadata against the assigned area.
9. The system sets the status to "Pending".
10. The system notifies the Supervisor and Admin.

Alternative Flow:

1. If the worker taps the floating "+" button, the worker selects an area from the map (assigned or unassigned) before capturing.
2. If the captured image does not contain valid metadata, the system warns the worker and allows recapture.
3. If there is no internet, the proof is cached locally and syncs when online.

Postcondition: The accomplishment proof is submitted and pending Supervisor/Admin approval.

3.17. UC-17 - Request Leave

Actor: Field Worker

Goal: To submit a leave request with a reason and optional proof image.

Precondition: The worker is logged in.

Main Flow:

1. The worker navigates to the Leave page.
2. The worker selects a leave type (Emergency, Sick, Vacation, Other).
3. The worker enters start date, end date, and reason (required).
4. The worker optionally attaches a proof image.
5. The worker submits the request.
6. The system sends a notification to the Supervisor (informed) and Admin (for approval).
7. The request status shows "Pending".

Alternative Flow:

1. If the worker submits a request without a reason, the system prevents submission and prompts for a reason.

Postcondition: The leave request is submitted and awaiting Admin approval.

3.18. UC-18 - View Performance Credits

Actor: Field Worker

Goal: To view personal performance credit score, color indicator, and credit history.

Precondition: The worker is logged in.

Main Flow:

1. The worker navigates to the Profile page.
2. The system displays the worker's star credit score (0-100) with color indicator: - 80-100: Green (Good) - 50-70: Yellow (Warning) - 10-40: Red (Critical).
3. The worker views the credit history: - Dates of deductions (absences) - Dates of additions (attendance, approved reports).
4. The worker views the attendance summary.

Alternative Flow:

1. If the credit score is Yellow (50-70), the system displays a warning message.

2. If the credit score is Red (10-40), the system displays a critical alert.

Postcondition: The worker can see their current performance status and history.

3.19. UC-19 - Offline Attendance Capture

Actor: Field Worker (System)

Goal: To log attendance when there is no internet connectivity.

Precondition: The worker has no internet connectivity; the worker is within the geofence.

Main Flow:

1. The worker opens the app (offline mode indicator shown).
2. The worker performs an attendance action.
3. The system captures a selfie with embedded metadata.
4. The system caches the attendance data locally (SQLite).
5. The system displays "Offline - Will sync when online" message.
6. When internet is restored, the system auto-syncs the cached data.
7. The system validates the metadata against the assigned area.

Alternative Flow:

1. If the worker attempts to submit multiple offline records without syncing, the system queues them in order of creation.

Postcondition: Attendance is captured offline and synced when connectivity is restored.

3.20. UC-20 - Offline Attendance Capture

Actor: System

Goal: To detect when a worker leaves their assigned geofence and notify relevant parties.

Precondition: The worker is logged in and has an active attendance session.

Main Flow:

1. The system monitors the worker's location continuously.
2. The worker leaves the assigned geofence.
3. The system detects the breach.
4. The system logs the breach event.
5. The system sends a warning notification to the worker.
6. The system sends a breach alert to the Supervisor.
7. The system sends a breach alert to the Admin.

Alternative Flow:

1. If the worker returns to the geofence within the allowed grace period, the system cancels the breach status.

Postcondition: The breach is logged; all relevant parties are notified.

3.21. UC-21 - Auto-Sync Offline Data

Actor: System

Goal: To automatically synchronize locally cached offline data when internet connectivity is restored.

Precondition: The device was offline; attendance and proof data were cached locally.

Main Flow:

1. The device detects internet connectivity restored.
2. The system checks the local cache (SQLite) for unsynced data.
3. The system uploads attendance records.
4. The system uploads accomplishment proofs.
5. The system validates metadata for each submission.
6. The system updates the status to "Synced".
7. The system notifies the user of successful sync.
8. If there are conflicts, the system alerts the Admin or Supervisor.

Alternative Flow:

1. If a sync conflict is detected (server record already exists), the system preserves the existing server record and logs the conflict.
2. If the sync fails due to server error, the system retries after a delay.

Postcondition: All offline data is synced to the server; no data loss; sync status is updated.

3.22. UC-22 - Auto-Sync Offline Data

Actor: System

Goal: To automatically generate and send an alert when a Field Worker has 3 or more consecutive absences.

Precondition: A Field Worker has 3 or more consecutive days of absence.

Main Flow:

1. The system detects the 3rd consecutive absence of a worker.
2. The system generates an alert.
3. The system adds the alert to the Admin's notification tab.
4. The system sends a notification to the Admin, Supervisor, and worker.
5. The Admin views the alert.
6. The Admin sends a warning notification to the worker.

Alternative Flow:

1. If the absence is due to an approved leave, the system does not count it as an absence.

Postcondition: All relevant parties are notified of the consecutive absence; the alert is available for Admin action.

4. USER STORIES

4.1. US-01 - Field Worker Registration

As an Administrator, I want to create employee accounts for field supervisors and field workers, so that they can access the system.

Acceptance Criteria:

● Administrator can create users and provide them usernames.

● The system displays an error message if registration fails due to invalid or duplicate information.

4.2. US-02 - Register Field Worker’s Face

As an Administrator, I want to register a worker’s face during account creation, so that the worker can use facial recognition for attendance.

Acceptance Criteria:

● Worker’s face successfully saved and registered.

● The system successfully extracts and stores the worker's facial data (face embedding).

4.3. US-03 - User Field Assignment

As an Administrator, I want to assign workers and supervisors to specific work areas so that attendance is location-restricted.

Acceptance Criteria:

● Administrator can assign a Field Supervisor to one or more work areas.

● The system displays a list of available work areas for assignment.

4.4. US-04 - Work Area Creation

As an Administrator, I want to create work areas using polygon geofencing on a map so that I can define irregular area boundaries.

Acceptance Criteria:

● Administrator can create a new work area using an interactive map.

● Administrator can draw a polygon to define the work area's boundary.

4.5. US-05 - Update Work Area

As an Administrator, I want to edit or delete work areas so that I can update boundaries when needed.

Acceptance Criteria:

● Administrator can edit the name of an existing work area.

● Administrator can modify the polygon boundary of an existing work area.

4.6. US-06 - Filtered List of Workers

As an Administrator, I want to view a list of all employees filtered by area so that I can manage them efficiently

Acceptance Criteria:

● Administrator can view a list of all registered employees.

● Administrator can filter the employee list by work area.

4.7. US-07 - View Worker Profile

As an Administrator, I want to view worker profiles with their performance and attendance history so that I can evaluate their work.

Acceptance Criteria:

● Administrator can view a worker's profile by selecting them from the employee list.

● The profile displays the worker's personal and employment information (e.g., name, role, assigned work area, supervisor).

4.8. US-08 - Leave Request Approval

As an Administrator, I want to approve or deny leave requests so that I can manage worker absences.

Acceptance Criteria:

● Administrator can view all submitted leave requests.

● Administrator can approve or deny a leave request.

4.9. US-09 - Validation of Accomplishment Reports

As an Administrator, I want to approve or deny accomplishment proofs so that I can validate work completion.

Acceptance Criteria:

● Administrator can view all submitted accomplishment reports.

● Administrator can approve or deny an accomplishment report.

4.10. US-10 - Report Generation

As an Administrator, I want to generate attendance and performance reports so that I can analyze workforce data

Acceptance Criteria:

● Administrator can generate attendance reports for a selected date range.

● Administrator can generate performance reports for a selected date range.

● Administrator can filter reports by work area, worker, or supervisor.

4.11. US-11 - Report Export

As an Administrator, I want to export reports to PDF or Excel so that I can share them with stakeholders.

Acceptance Criteria:

● Administrator can export generated reports in PDF/Excel format.

● The exported report includes relevant details such as the report title, date generated, and selected filters.

4.12. US-12 - Notification and Announcement Management

As an Administrator, I want to send notifications/announcements to all users, specific roles, or specific users so that I can communicate effectively.

Acceptance Criteria:

● Administrator can create a new announcement or notification.

● Administrator can send the announcement to all users, selected roles, or specific users.

4.13. US-13 - Private Notifications

As an Administrator, I want to send confidential notifications so that sensitive feedback is private.

Acceptance Criteria:

● Administrator can compose a private notification.

● The system delivers the notification only to the selected recipient.

4.14. US-14 - Consecutive Absence Alerts

As an Administrator, I want to receive alerts for consecutive absences (3+ days) so that I can take action.

Acceptance Criteria:

● The system automatically detects workers who have been absent for three (3) or more consecutive working days.

● The system sends an alert to the administrator when the absence threshold is reached.

4.15. US-15 - Field Supervisor Dashboard

As a Field Supervisor, I want to view my assigned areas on a dashboard so that I know where to monitor.

Acceptance Criteria:

● The dashboard displays all work areas assigned to the supervisor.

● The dashboard displays a summary of attendance for the supervisor's assigned work areas.

4.16. US-16 - View Attendance Summary

As a Field Supervisor, I want to view attendance summary of workers assigned to my areas so that I can monitor their presence.

Acceptance Criteria:

● Field Supervisor can view the attendance summary of workers assigned to their work areas.

● The system displays each worker's attendance status (e.g., Present, Absent, Late, On Leave).

4.17. US-17 - Accomplishment Validation in Assigned Area

As a Field Supervisor, I want to approve or deny accomplishment proofs for my assigned areas so that I can validate work completion.

Acceptance Criteria:

● Field Supervisor can view accomplishment reports submitted by workers assigned to their work areas.

● The system displays the report details, including the worker, submission date, description, work area, and attached proof.

4.18. US-18 - View Leave Requests in Assigned Area

As a Field Supervisor, I want to view leave requests from workers in my assigned areas so that I am informed.

Acceptance Criteria:

● Field Supervisor can view leave requests submitted by workers assigned to their work areas.

● The system displays only leave requests from workers within the supervisor's assigned work areas.

4.19. US-19 - Report Worker Issues

As a Field Supervisor, I want to report worker issues to the Admin so that they can take action.

Acceptance Criteria:

● The supervisor can enter the issue details and requires to attach supporting evidence (e.g., photos or documents).

● The Administrator is notified when a new worker issue report is submitted.

4.20. US-20 - Report Generation on Assigned Area

As a Field Supervisor, I want to generate reports for my assigned areas so that I can analyze workforce data

Acceptance Criteria:

● Field Supervisor can generate attendance and performance reports for their assigned work areas.

● The system allows the supervisor to specify a date range for the report.

4.21. US-21 - Report Export on Assigned Area

As a Field Supervisor, I want to export reports to PDF or Excel so that I can share them with Admin.

Acceptance Criteria:

● Field Supervisor can export generated reports in PDF/Excel format.

● The exported report contains only data from the supervisor's assigned work areas.

4.22. US-22 - Leave Request Notification

As a Field Supervisor, I want to receive leave request notifications so that I am informed of worker absences.

Acceptance Criteria:

● The system notifies the Field Supervisor when a worker in their assigned work area submits a leave request.

● Field Supervisor can open the notification to view the leave request details.

4.23. US-23 - Persistent Login

As a Field Worker, I want to stay logged in permanently after first login so that I don't need to log in every day.

Acceptance Criteria:

● Field Worker remains logged in after successfully completing the initial login.

● The system automatically restores the user's session when the application is reopened.

4.23. US-23 - Automatic Image Metadata Embedding

As a Field Worker, I want the image to automatically have date, time, and location embedded so that my proof is verifiable.

Acceptance Criteria:

● The system automatically embeds the current date, time and location on every proof image captured within the application.

● The system automatically burns the date, time, and location on the image itself.

4.24. US-24 - Submit Accomplishment Report

As a Field Worker, I want to submit accomplishment reports for my assigned area by default so that it's quick and easy.

Acceptance Criteria:

● The system automatically selects the worker's assigned work area when creating an accomplishment report.

● The worker can enter a description and attach one or more proof images before submitting the report.

4.25. US-25 - Submit Additional Accomplishment Report

As a Field Worker, I want to submit additional accomplishment reports for any area I worked on so that my extra work is recorded.

Acceptance Criteria:

● Field Worker can create an additional accomplishment report for a work area other than their assigned area.

● The system allows the worker to select the work area for the additional accomplishment report.

4.26. US-26 - Leave Request

As a Field Worker, I want to request leave with a reason and optional proof image so that I can be excused from work.

Acceptance Criteria:

● Field Worker can create a leave request by specifying the leave dates, leave type, and reason.

● The worker can optionally attach a supporting proof image or document to the leave request.

4.27. US-27 - Offline Mode

As a Field Worker, I want to use the app offline so that I can log attendance and submit proofs even without internet,

Acceptance Criteria:

● Field Worker can record attendance without an active internet connection.

● Field Worker can capture and submit accomplishment reports while offline.

● The system securely stores offline attendance records and reports on the device.

4.28. US-28 - Automatic Data Synchronization

As a Field Worker, I want my offline data to auto-sync when internet is available so that I don't lose anything.

Acceptance Criteria:

● The system automatically detects when an internet connection becomes available.

● The system automatically synchronizes all pending offline attendance records, accomplishment reports, and leave requests.

4.29. US-29 - Camera Selection for Proof Images

As a Field Worker, I want to use the front or back camera for proof images so that I can work around hardware issues.

Acceptance Criteria:

● Field Worker can switch between the front and back camera before capturing a proof image.

● The system remembers the last selected camera during the current session.

4.30. US-30 - Simple and Intuitive User Interface

As a Field Worker, I want to have a minimal and clean user interface so that the application is easy to understand and use.

Acceptance Criteria:

● Navigation is simple and allows workers to access common tasks (e.g., attendance, accomplishment reports, leave requests) within a few taps.

● Buttons, icons, and labels are clearly visible and easy to understand.

● The interface is responsive and usable on Android 7.0+ devices with different screen sizes.

5. FUNCTIONAL REQUIREMENTS

5.1. Functional Requirements Table

Requirement ID	Requirement	Priority	Source
FR-01	The system shall allow Admin and Field Supervisor to log in using Username, Password, and MFA (OTP)	High	UC-01, UC-10
FR-02	The system shall allow Field Workers to log in via QR code scanning on first use	High	UC-14
FR-03	The system shall store Field Worker sessions persistently after first login	High	UC-14
FR-04	The system shall support facial recognition as an authentication factor for attendance actions	High	UC-15
FR-05	The system shall support three user roles: Admin, Field Supervisor, Field Worker	High	UC-01
FR-06	The system shall restrict access to features based on user role	High	UC-01, UC-03, UC-04, UC-10
FR-07	The system shall allow Admin to create, update, delete, and view all user accounts	High	UC-01
FR-08	The system shall allow Admin to assign roles to users	High	UC-01
FR-09	The system shall allow Admin to view all user profiles	High	UC-01
FR-10	The system shall allow Admin to assign multiple work areas to a Field Supervisor	High	UC-02
FR-11	The system shall allow Admin to assign multiple Field Workers to a work area	High	UC-02

FR-12	The system shall allow Admin to set work schedules per worker	Medium	UC-01
FR-13	The system shall allow Admin to view worker performance metrics	High	UC-05
FR-14	The system shall allow Admin to create,edit,delete work areas with defined geofence boundaries	High	UC-02
FR-15	The system shall require the worker to be within the assigned geofence for all attendance actions	High	UC-15, UC-20
FR-16	The system shall record GPS coordinates for all attendance actions	High	UC-15
FR-17	The system shall allow workers to view their attendance history	Medium	UC-15
FR-18	The system shall allow Admin to view attendance records for all areas	High	UC-05
FR-19	The system shall allow Supervisor to view attendance records for assigned areas only	High	UC-10
FR-20	The system shall allow Field Workers to submit accomplishment reports per task completion	High	UC-16
FR-21	The system shall allow Supervisor to accept or reject accomplishment proofs for assigned areas only	High	UC-04
FR-22	The system shall allow workers to resubmit rejected proofs	Medium	UC-16
FR-23	The system shall allow Field Workers to submit leave requests anytime	High	UC-17
FR-24	The system shall allow Field Supervisors to view leave requests from assigned workers	High	UC-13

FR-25	The system shall allow Admin to send warning notifications to workers	High	UC-06,UC-09
FR-26	The system shall allow Admin to generate attendance reports for all areas	High	UC-05
FR-27	The system shall allow Supervisor to generate reports for assigned areas only	High	UC-12
FR-28	The system shall display real-time worker locations on a map	High	UC-08
FR-29	The system shall display assigned work zones (geofences) on the map	High	UC-08
FR-30	The system shall allow Admin to view the GIS map	High	UC-08
FR-31	The system shall allow Supervisor to view the GIS map for assigned areas only	High	UC-08
FR-32	The system shall allow Admin to view all worker reports	High	UC-05
FR-33	The system shall allow attendance logging without internet connectivity	High	UC-19
FR-34	The system shall allow image proof capture without internet connectivity	High	UC-19
FR-35	The system shall automatically sync offline data when internet is restored	High	UC-21
FR-36	The system shall display behavior or credit score of a field worker	High	UC-18

6. NON-FUNCTIONAL REQUIREMENTS

6.1. Non-Functional Requirements Table

The non-functional requirements below are organized according to the ISO/IEC 25010 software product quality characteristics that will later serve as the basis for the system's evaluation.

Req. ID	Category	Requirement Statement	Measurement / Target
NFR-01	Performance	The system shall log attendance actions within acceptable time	< 3 seconds response time
NFR-02	Performance	The system shall load the GIS map within acceptable time	< 5 seconds
NFR-03	Performance	The system shall launch the mobile application within acceptable time	< 5 seconds
NFR-04	Performance	The system shall process facial recognition within acceptable time	< 3 seconds
NFR-05	Performance	The system shall deliver push notifications within acceptable time	< 5 seconds
NFR-06	Performance	The system shall respond to simple API queries within acceptable time	< 500ms
NFR-07	Performance	The system shall respond to complex API queries within acceptable time	< 2 seconds
NFR-08	Resource Utilization	The system shall support concurrent users without performance degradation	50+ concurrent users
NFR-09	Resource Utilization	The system shall maintain consistent frame rate during operation	60 fps minimum
NFR-10	Resource Utilization	The system shall optimize battery usage for field workers	Battery drain < 20% per 4-hour shift

NFR-11	Confidentiality	The system shall encrypt all data transmission between client and server	HTTPS / TLS 1.2+
NFR-12	Confidentiality	The system shall encrypt sensitive user data at rest	AES-256 encryption
NFR-13	Confidentiality	The system shall store facial recognition data locally on devices only	Not stored in cloud
NFR-14	Integrity	The system shall store passwords using strong cryptographic hashing	BCrypt or equivalent
NFR-15	Integrity	The system shall use JWT tokens for API authentication	JWT with secure signing
NFR-16	Integrity	The system shall generate unique and secure QR codes for worker login	Cryptographically random QR codes
NFR-17	Authentication	The system shall implement Multi-Factor Authentication	Password + OTP + Face
NFR-18	Authorization	The system shall enforce Role-Based Access Control	Restrict actions based on user role
NFR-19	Authorization	The system shall implement session timeout after inactivity	30 minutes of inactivity
NFR-20	Non-Repudiation	The system shall maintain audit logs for all Admin actions	Logs stored for 6 months minimum
NFR-21	Compliance	The system shall comply with Philippine data privacy regulations	Data Privacy Act of 2012 (RA 10173)
NFR-22	Appropriateness Recognizability	The system shall be intuitive for users with minimal training	80% of users complete tasks without assistance
NFR-23	Learnability	The system shall include an in-app tutorial for first-time users	Users complete tutorial in < 5 minutes
NFR-24	Operability	The system shall be designed	Mobile-first

		mobile-first for field worker use	responsive design
NFR-25	Operability	The system shall have large touch targets for field use on mobile	Touch targets > 44dp
NFR-26	User Error Protection	The system shall provide clear visual feedback for user actions	Visual confirmation within 1 second
NFR-27	User Interface Aesthetics	The system shall support portrait mode only for mobile	Portrait mode locked
NFR-28	Accessibility	The system shall support English and Filipino languages	Language toggle available
NFR-29	Accessibility	The system shall provide clear error messages for failed operations	User-friendly error descriptions
NFR-30	Maturity	The system shall maintain high availability during working hours	99% uptime
NFR-31	Fault Tolerance	The system shall handle network interruptions without data loss	No data loss on network failure
NFR-32	Fault Tolerance	The system shall maintain core features functionality offline	Attendance + proof capture work offline
NFR-33	Recoverability	The system shall automatically sync offline data when connectivity restores	Sync completes within 1 minute
NFR-34	Recoverability	The system shall ensure no duplicate records during sync	No duplicate records created
NFR-35	Capacity	The system shall support additional users without performance degradation	Support 100+ total users
NFR-36	Capacity	The system shall support addition of new work areas without performance impact	Unlimited work areas
NFR-37	Capacity	The database shall support efficient spatial queries	PostGIS indexing optimized

NFR-38	Modularity	The system shall use modular architecture for easy maintenance	Separation of concerns in code
NFR-39	Reusability	The system shall allow future feature additions	Extensible architecture
NFR-40	Analyzability	The system shall log system errors for debugging	Error logs stored for 30 days
NFR-41	Modifiability	The system shall support easy backup and restore of data	Weekly automated backups
NFR-42	Adaptability	The mobile app shall support multiple Android versions	Android 7.0 and above
NFR-43	Adaptability	The web dashboard shall support modern browsers	Chrome, Firefox, Edge, Safari
NFR-44	Installability	The mobile app shall be deployable from standard app stores	Google Play Store compatible
NFR-45	Co-existence	The mobile app shall work with devices with camera and GPS	GPS + Camera required
NFR-46	Co-existence	The mobile app shall support both front and back cameras	Both cameras functional
NFR-47	Co-existence	The system shall support devices with varying screen sizes	360dp x 800dp minimum resolution
NFR-48	Co-existence	The system shall support devices with 2GB+ RAM	2GB minimum RAM
NFR-49	Accuracy	The system shall detect and resolve sync conflicts automatically	Conflicts resolved with server priority
NFR-50	Accuracy	The system shall validate metadata against assigned area during sync	Metadata validation performed
NFR-51	Completeness	The system shall sync all offline data	100% data synced

		when internet is restored	
NFR-52	Timeliness	Sync status shall be visible to users	Sync status indicator displayed
NFR-53	Durability	Images captured offline shall retain metadata and visual overlay	Metadata preserved after sync

7. SYSTEM ARCHITECTURE

7.1. System Architecture

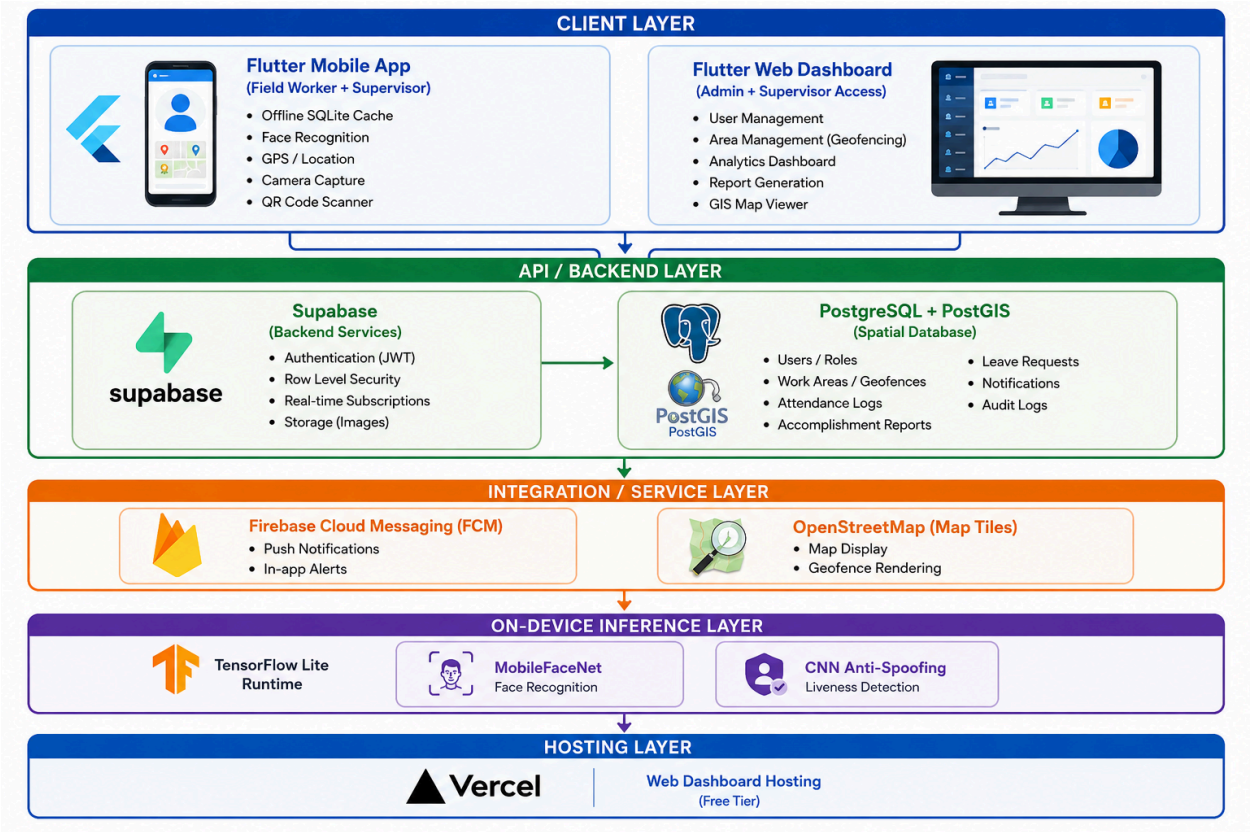
The proposed system employs a three-tiered client-server architecture that consists of a Flutter presentation layer, a Supabase application layer, and a PostgreSQL + PostGIS data layer. The presentation layer handles all user interface-related responsibilities such as user authentication, input collection, and information display for the Admin, Field Supervisor, and Field Worker user roles. The mobile application offers a mobile-first user experience to the Field Worker while the web dashboard provides a unified view of the system to the Admin and Field Supervisor. The application layer is primarily concerned with the execution of the system's core features, workflows, and business logic. It handles RESTful API request handling and edge function execution among other responsibilities such as user verification and validation. Lastly, the data layer stores and manages the system's data, including user profiles, face embeddings, geo-fencing polygons, attendance records, accomplishment proofs, and leaves approvals data. The system uses TensorFlow Lite for on-device facial recognition, Firebase Cloud Messaging for push notification handling, and the OpenStreetMap tile service via the flutter_map package for map visualization. The architecture ensures modularity, maintainability, and scalability while supporting the development of offline-first applications and ensuring data and application-level security.

7.2. Technology Stack

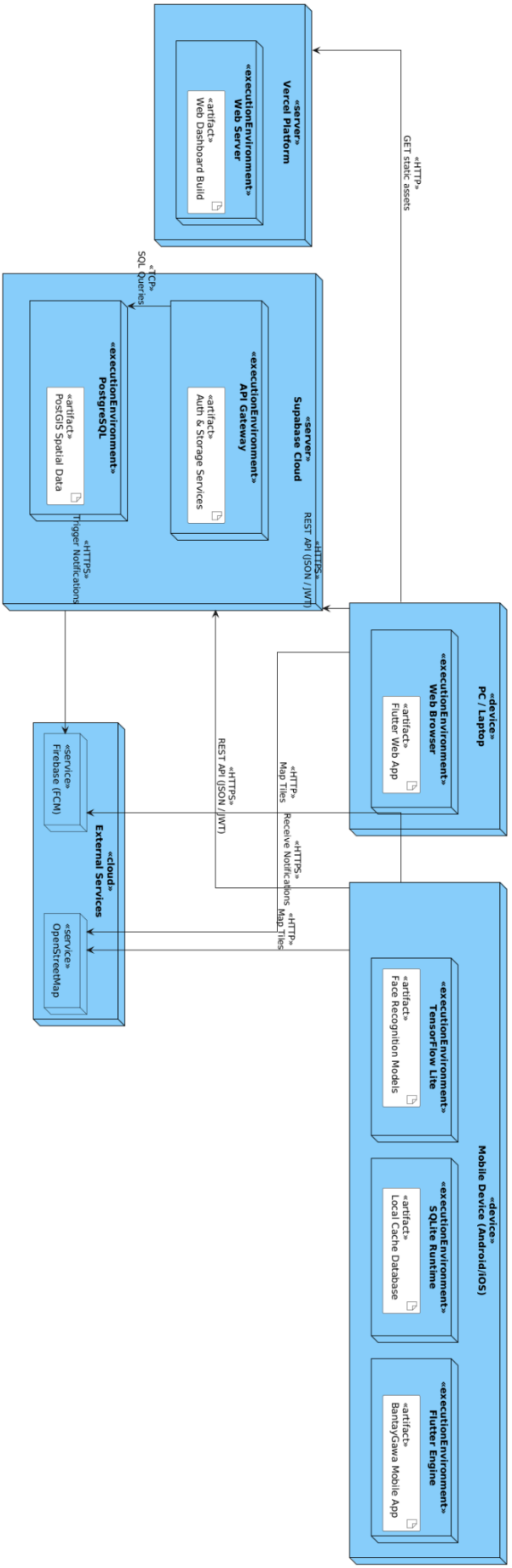
Layer	Technology	Purpose
Frontend (Mobile App)	Flutter	Cross-platform mobile application framework for Android and iOS
	flutter_map	GIS/Map rendering library for OpenStreetMap integration
	SQLite	Local offline database for caching and sync
	TensorFlow Lite	On-device facial recognition inference runtime
Frontend (Web Dashboard)	Flutter Web	Cross-platform web application for Admin and Supervisor dashboards
	flutter_map	GIS/Map rendering for web dashboard
Backend	Supabase	Backend-as-a-Service providing authentication, database, real-time subscriptions, and file storage

	JWT (JSON Web Tokens)	API authentication and session management
	Firebase Cloud Messaging (FCM)	Push notification delivery service
	MobileFaceNet	Facial recognition model for face detection and verification (on-device)
	CNN Anti-Spoofing	Liveness detection model to prevent photo/video spoofing (on-device)
	Vercel	Web dashboard hosting service (free tier)
Database Layer	PostgreSQL	Primary relational database management system
	PostGIS	Spatial database extension for PostgreSQL (geofence storage and queries)
	Supabase Storage	Cloud file storage for proof images and profile photos

7.3. System Architecture Diagram



7.4. Deployment Diagram



8. Interface Requirements

8.1 User Interface Requirements

The system shall provide the GUI for the following users:

Admin Dashboard (Web + Mobile):

- Login
- Dashboard
- Employee Management
- Area Management
- Attendance Management
- Leave Management
- Reports
- Notifications
- Profile

Field Supervisor (Web + Mobile):

- Login
- Dashboard
- Worker Management
- Area Management
- Attendance
- Accomplishment Proofs
- Leave Management
- Reports
- Profile

Field Worker (Mobile):

- First-Time Login/Scan ID
- Dashboard
- Attendance
- Area Map
- Accomplishment

- Leave
- Notifications
- Profile
- Report Worker

8.2. Software Interface Requirements

The Software Interface Requirements describes how every software application communicates in the proposed system.

Software Applications	Purpose
Flutter	For developing both the mobile app and web application.
Supabase	Managing the backend, authentication, storage, and database connection.
PostGIS	Stores and processes location and geofence data.
Mobilefacenet	For verifying the user's identity through facial recognition.
CNN Anti Spoofing	For detecting fake faces to prevent attendance fraud.
Flutter map	Displays interactive maps in the application.
Open Street Map	Provides map data for location and geofencing features.
Firebase Cloud Messaging	Sends push notifications to users. (Does not send email notifications.)
Vercel Hosting	Hosts the web dashboard online.
JWT	Secures user authentication and API access.

8.3 Hardware Interface Requirements

The Hardware interface requirements specifies the hardware/physical devices used and needed to interact with the software requirements.

Hardware	Purpose
WebCam	For face recognition and for employee registration.
Computer/Laptop	For monitoring reports, employees and for attendance checking.
Phone	For user interaction.

8.4. External Interface Requirements

External Interfaces describes how proposed systems exchange information with external systems, organization, and services.

External Interface	Purpose
Third Party APIs	Integrates with external APIs, platforms, and services.

8.5. API Integration

This Application Programming Interface (API) describes the external and backend service APIs integrated into the proposed system to handle authentication, real-time messaging, spatial querying, cloud storage, and push notifications.

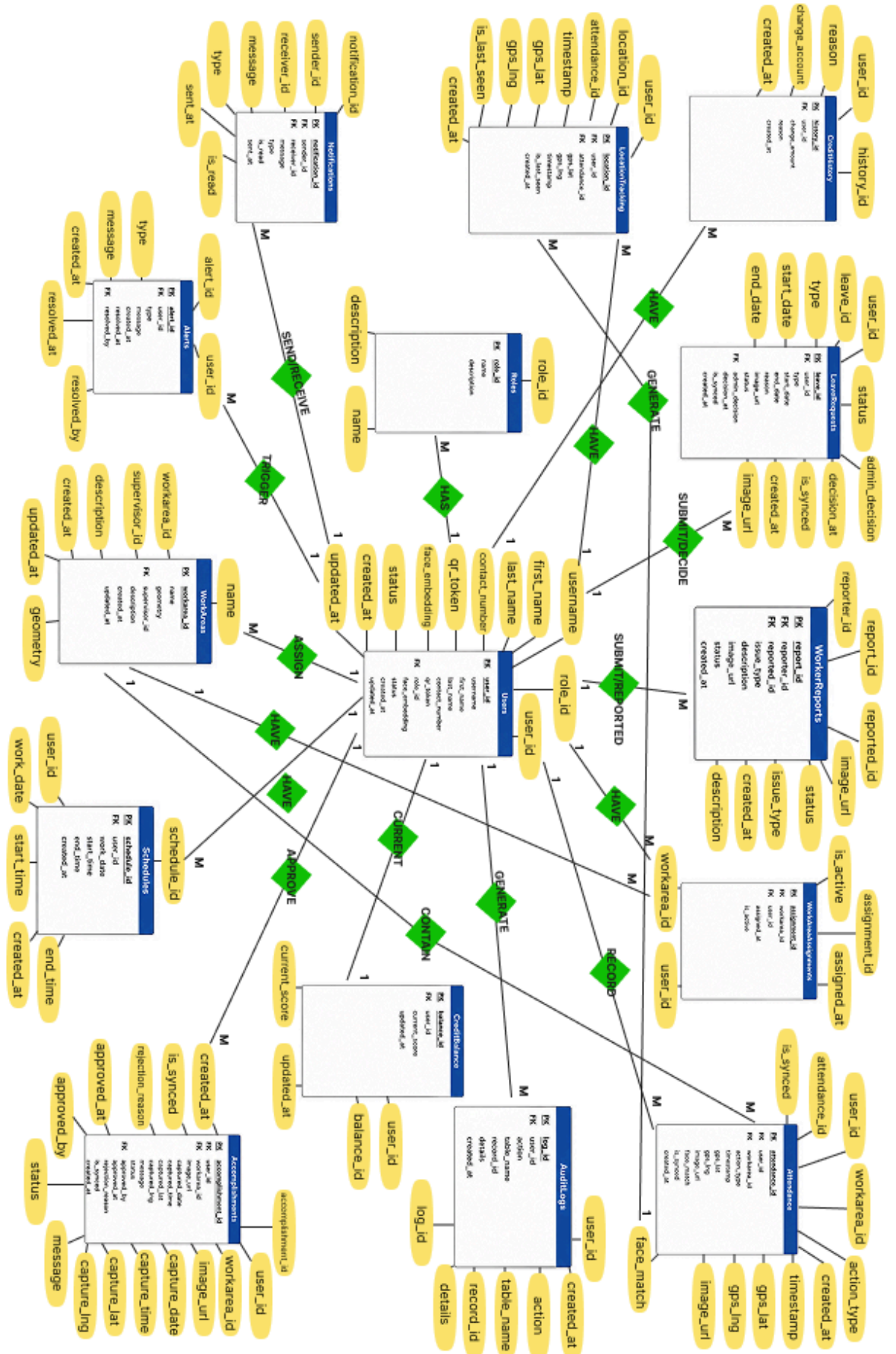
- Supabase REST API
- Firebase FCM API
- PostGIS Spatial Queries
- JWT Authentication
- Supabase Storage
- Supabase Realtime

9. Data Requirements

9.1 Database Overview

The BantayGawaMobile database stores and manages all system data, including user accounts, work areas, attendance records, accomplishment proofs, leave requests, credit scores, notifications, alerts, and system logs. The database uses PostgreSQL, a reliable relational database system, with additional support for location-based data such as work area boundaries and GPS coordinates. Data is organized into fifteen main tables that cover user management (Users, Roles, Schedules), workforce operations (WorkAreas, WorkAreaAssignments, Attendance, LocationTracking, Accomplishments, CreditBalance, CreditHistory), and workflow management (LeaveRequests, Notifications, Alerts, WorkerReports, AuditLogs). When workers are offline, their attendance and proof submissions are saved locally on their mobile device using SQLite and automatically uploaded to the cloud database once internet connectivity is restored. Images submitted as proof of work are stored securely in the cloud with their metadata (date, time, location) extracted and saved in the database for easy searching and reporting. The database uses security rules to ensure that users can only access data they are authorized to view—workers see only their own records, supervisors see records for their assigned areas, and the admin has full access. This design ensures that the system remains reliable and accurate whether workers are online or offline.

Entity Relationship Diagram - ERD



9.2 Entities and Attributes

An Entity represents a person, events, place, real world object, or concept which is the data being stored.

USERS	
user_id	A unique identifier for each user (linked to Supabase auth)
username	A unique username for login
first_name	The employee's first name
last_name	The employee's last name
contact_number	The employee's contact information
qr_token	A unique, regeneratable token for QR code login
role_id	A foreign key referencing the Roles table
face_embedding	A facial recognition feature vector (stored during registration)
status	The account status (active, inactive, pending)
created_at	The account creation timestamp
updated_at	The last update timestamp

ROLES	
role_id	A unique identifier for each role
name	The role name (e.g., Admin, Field Supervisor, Field Worker)
description	A brief description of role permissions

WORK_AREAS	
workarea_id	A unique identifier for each work area
name	The name of the work area
geometry	A PostGIS polygon defining the geofence boundary
supervisor_id	A foreign key referencing the supervisor assigned to the area
description	An optional description of the work area
created_at	The creation timestamp
updated_at	The last update timestamp

WORK_AREA_ASSIGNMENTS	
assignment_id	A unique identifier for each assignment
workarea_id	A foreign key referencing the WorkAreas table
user_id	A foreign key referencing the Field Worker assigned
assigned_at	The date and time of the assignment
is_active	A flag indicating if the assignment is currently active

ATTENDANCE	
attendance_id	A unique identifier for each attendance record
user_id	A foreign key referencing the Users table
workarea_id	A foreign key referencing the WorkAreas table
action_type	The type of action (time-in, break-in,

	break-out, time-out)
timestamp	The exact time of the attendance action
gps_lat	The GPS latitude at the attendance time
gps_lng	The GPS longitude at the attendance time
image_url	A URL to the selfie proof image
face_match	A boolean indicating if facial recognition was successful
is_synced	A boolean indicating the offline sync status
created_at	The record creation timestamp

LOCATION_TRACKING	
location_id	A unique identifier for each location record
user_id	A foreign key referencing the Users table
attendance_id	A foreign key referencing the active Attendance session
gps_lat	The GPS latitude at the tracking point
gps_lng	The GPS longitude at the tracking point
timestamp	The exact time the location was captured
is_last_seen	Marks the last location before leaving a geofence (for breach detection)
created_at	The record creation timestamp

ACCOMPLISHMENTS	
accomplishment_id	A unique identifier for each accomplishment report
user_id	A foreign key referencing the submitting Field Worker
workarea_id	A foreign key referencing the WorkAreas

	table
image_url	A URL to the work proof image
captured_date	The date the image was captured (extracted from metadata)
captured_time	The time the image was captured (extracted from metadata)
captured_lat	The GPS latitude extracted from image metadata
captured_lng	The GPS longitude extracted from image metadata
message	An optional message provided by the worker
status	The report status (pending, approved, denied)
approved_by	A foreign key referencing the Admin or Supervisor who approved it
approved_at	The timestamp of approval
rejection_reason	The reason provided if the report was denied
is_synced	A boolean indicating the offline sync status
created_at	The submission timestamp

LEAVE_REQUESTS	
leave_id	A unique identifier for each leave request
user_id	A foreign key referencing the requesting worker
type	The type of leave (emergency, sick, vacation, other)
start_date	The starting date of the leave
end_date	The ending date of the leave

reason	The stated reason for the leave
image_url	A URL to an optional proof image (e.g., medical certificate)
status	The request status (pending, approved, denied)
admin_decision	A foreign key referencing the Admin who made the decision
decision_at	The timestamp of the decision
is_synced	A boolean indicating the offline sync status
created_at	The submission timestamp

CREDIT_BALANCE	
balance_id	A unique identifier for the credit balance
user_id	A foreign key referencing the Users table
current_score	The current credit score
updated_at	The last update timestamp

CREDIT_HISTORY	
history_id	A unique identifier for each credit transaction
user_id	A foreign key referencing the Users table
change_amount	The number of points added or deducted
reason	The reason for the change in points
created_at	The transaction timestamp

NOTIFICATIONS

notification_id	A unique identifier for each notification
sender_id	A foreign key referencing the sender
receiver_id	A foreign key referencing the receiver
message	The content of the notification
type	The notification type (announcement, confidential, alert)
is_read	The read status of the notification
sent_at	The timestamp when the notification was sent

ALERTS	
alert_id	A unique identifier for each system alert
user_id	A foreign key referencing the affected worker
type	The alert type
message	The alert message
created_at	The alert generation timestamp
resolved_at	The timestamp when the alert was resolved
resolved_by	A foreign key referencing the Admin who resolved the alert

WORKER_REPORTS	
report_id	A unique identifier for each worker incident report
reporter_id	A foreign key referencing the user who submitted the report
reported_id	A foreign key referencing the user being reported

issue_type	The type of issue reported
description	A detailed description of the issue
image_url	A URL to an optional proof image
status	The report status
created_at	The report submission timestamp

SCHEDULES	
schedule_id	A unique identifier for each shift schedule
user_id	A foreign key referencing the Users table
start_time	The starting time of the shift
end_time	The ending time of the shift
days	A comma-separated list of working days
created_at	The schedule creation timestamp

AUDIT_LOGS	
log_id	A unique identifier for each audit log entry
user_id	A foreign key referencing the user who performed the action
action	The type of action
table_name	The name of the affected database table
record_id	The UUID of the affected record
old_values	The previous values of the record
new_values	The new values of the record
created_at	The timestamp of the action

9.3 Relationship Between Entities

Relationship describes how entities interact within the database.

1. Users and Roles (Many-to-One)
    - Each user is assigned exactly one role (e.g., Admin, Supervisor, Worker).
    - A specific role can be held by many users.
    - Foreign Key Constraint: Users.role_id references Roles.role_id

2. Users and Work Areas [Supervisor] (One-to-Many)
    - A Field Supervisor can be assigned to oversee multiple work areas.
    - A single work area has exactly one designated Field Supervisor.
    - Foreign Key Constraint: WorkAreas.supervisor_id references Users.user_id

3. Users and Work Area Assignments (One-to-Many
    - A Field Worker can have multiple assignment records over time (to track historical reassignments).
    - A specific assignment record belongs to exactly one user.
    - Foreign Key Constraint: WorkAreaAssignments.user_id references Users.user_id

4. Work Areas and Work Area Assignments (One-to-Many)
    - A work area can have many workers assigned to it simultaneously.
    - A specific assignment record belongs to exactly one work area.
    - Foreign Key Constraint: WorkAreaAssignments.workarea_id references WorkAreas.workarea_id

5. Users and Attendance (One-to-Many)
    - A user can have many attendance records throughout their employment.
    - An attendance record belongs to exactly one user.
    - Foreign Key Constraint: Attendance.user_id references Users.user_id

6. Work Areas and Attendance (One-to-Many)
    - A work area can contain many attendance records logged by various workers.
    - An attendance record is captured inside exactly one work area.
    - Foreign Key Constraint: Attendance.workarea_id references WorkAreas.workarea_id

7. Users and Location Tracking (One-to-Many)
    - A user can have many location tracking coordinate records generated over time.
    - A location tracking record belongs to exactly one user.
    - Foreign Key Constraint: LocationTracking.user_id references Users.user_id

8. Attendance and Location Tracking (One-to-Many)
    - An active attendance session (e.g., between time-in and time-out) generates many continuous location tracking points.
    - A location tracking record is tied to exactly one active attendance session.
    - Foreign Key Constraint: LocationTracking.attendance_id references Attendance.attendance_id

9. Users and Accomplishments (One-to-Many)
    - A Field Worker can submit multiple accomplishment proof reports.
    - An accomplishment report is submitted by exactly one user.
    - Foreign Key Constraint: Accomplishments.user_id references Users.user_id

10. Work Areas and Accomplishments (One-to-Many)
    - A work area can contain many submitted accomplishment reports.
    - An accomplishment report is submitted within exactly one work area.
    - Foreign Key Constraint: Accomplishments.workarea_id references WorkAreas.workarea_id

11. Users and Accomplishments [Approval] (One-to-Many)
    - An Admin or Field Supervisor can approve many accomplishment reports.
    - An accomplishment report is approved by exactly one Admin or Supervisor.
    - Foreign Key Constraint: Accomplishments.approved_by references Users.user_id

12. Users and Leave Requests (One-to-Many)
    - A user can submit multiple leave requests.
    - A leave request belongs to exactly one user.
    - Foreign Key Constraint: LeaveRequests.user_id references Users.user_id

13. Users and Leave Requests [Decision] (One-to-Many)

- An Admin can decide on (approve or deny) many leave requests.
- A leave request is decided by exactly one specific Admin.
- Foreign Key Constraint: LeaveRequests.admin_decision references Users.user_id

14. Users and Credit Balance (One-to-One)
    - Each user has exactly one active current credit balance/score.
    - Foreign Key Constraint: CreditBalance.user_id references Users.user_id

15. Users and Credit History (One-to-Many)
    - A user can have many credit transaction history records (points added or deducted).
    - A credit history transaction belongs to exactly one user.
    - Foreign Key Constraint: CreditHistory.user_id references Users.user_id

16. Users and Notifications [Sender] (One-to-Many)
    - A user (typically an Admin) can send many notifications.
    - A specific notification is sent by exactly one user.
    - Foreign Key Constraint: Notifications.sender_id references Users.user_id

17. Users and Notifications [Receiver] (One-to-Many)
    - A user can receive many notifications.
    - A specific notification is received by exactly one user.
    - Foreign Key Constraint: Notifications.receiver_id references Users.user_id

18. Users and Alerts (One-to-Many)
    - A user's actions (e.g., consecutive absences) can trigger multiple system alerts.
    - An alert is triggered for exactly one specific user.
    - Foreign Key Constraint: Alerts.user_id references Users.user_id

19. Users and Alerts [Resolution] (One-to-Many)
    - An Admin can resolve many system alerts.
    - An alert is resolved by exactly one specific Admin.
    - Foreign Key Constraint: Alerts.resolved_by references Users.user_id

20. Users and Worker Reports [Reporter] (One-to-Many)
    - A user can submit many incident reports about other workers.
    - An incident report is submitted by exactly one user.
    - Foreign Key Constraint: WorkerReports.reporter_id references Users.user_id

21. Users and Worker Reports [Reported] (One-to-Many)
    - A user can be reported multiple times by different peers or supervisors.
    - An incident report targets exactly one specific user.
    - Foreign Key Constraint: WorkerReports.reported_id references Users.user_id

22. Users and Schedules (One-to-Many)
    - A user can have multiple shift schedules over time (for historical tracking of shift changes).
    - A schedule record belongs to exactly one user.

- Foreign Key Constraint: Schedules.user_id references Users.user_id

23. Users and Audit Logs (One-to-Many)
    - A user (Admin or Supervisor) can generate many audit log entries based on their actions within the system.
    - An audit log entry records the action of exactly one user.
    - Foreign Key Constraint: AuditLogs.user_id references Users.user_id

9.4 Database Normalization

The database for the BantayGawaMobile system has been designed and normalized up to the Third Normal Form (3NF). Database normalization is a systematic approach to organizing tables and columns to reduce data redundancy, improve data integrity, and prevent insert, update, and delete anomalies.

The system adheres to the following normalization stages:

1. First Normal Form (1NF) - Elimination of Repeating Groups To satisfy 1NF, every attribute in the database holds only atomic (indivisible) values, and each record is unique. There are no repeating groups or arrays of data stored within a single column.
    - Application in the System: Instead of storing a comma-separated list of assigned work areas directly inside the Users table, a dedicated junction table (WorkAreaAssignments) was created. This ensures that every assignment is a distinct, atomic record mapping one user to one work area at a time.

2. Second Normal Form (2NF) - Elimination of Partial Dependencies To satisfy 2NF, the database must first be in 1NF, and all non-key attributes must be fully functionally dependent on the primary key.

- Application in the System: The system inherently satisfies 2NF by utilizing single-column, unique surrogate keys (UUIDs) such as user_id, attendance_id, and workarea_id as the Primary Keys for every table. Because composite primary keys are avoided, there are no partial dependencies; every attribute in a table relies entirely on the single primary key of that table.

3. Third Normal Form (3NF) - Elimination of Transitive Dependencies To satisfy 3NF, the database must be in 2NF, and there must be no transitive dependencies. This means that a non-key attribute must not depend on another non-key attribute; it must depend only on the primary key.
    - Application in the System: The system eliminates transitive dependencies through proper table relationships and Foreign Keys. For example, rather than storing a user's role name (e.g., "Field Supervisor") and role description directly inside the Users table, which would create a transitive dependency on the role_id, this data is separated into an independent Roles table. The Users table only stores the role_id as a Foreign Key. This ensures that if a role's description changes, it only needs to be updated in one place, preventing data inconsistency.

9.5 Data Dictionary

The Data Dictionary defines all fields stored in the system's database. It details field, data types,length, and field descriptions. This serves as a guide for developers, database administrators, and testers to ensure data consistency.

USERS			
Field	**Data Type**	**Length**	**Description**
user_id	UUID	36 chars	Primary Key

username	VARCHAR	50	Unique username for login
first_name	VARCHAR	100	The employee's first name
last_name	VARCHAR	100	The employee's last name
contact_number	VARCHAR	20	The employee's contact number
qr_token	VARCHAR	100	Unique, regeneratable token for QR code login
role_id	UUID	36 chars	Foreign Key referencing the ROLES table
face_embedding	TEXT	N/A	Facial recognition feature vector string
status	VARCHAR	20	Account status (active, inactive, pending)
created_at	TIMESTAMP	8 bytes	Timestamp when the account was created
updated_at	TIMESTAMP	8 bytes	Timestamp when the account was last updated

ROLES			
Field	**Data Type**	**Length**	**Description**
role_id	UUID	36 chars	Primary Key
name	VARCHAR	50	The role name (Admin, Supervisor, Worker)
description	TEXT	N/A	A brief description of role permissions

WORK_AREAS			
Field	**Data Type**	**Length**	**Description**
workarea_id	UUID	36 chars	Primary Key
name	VARCHAR	100	The name of the work area
geometry	GEOMETRY	N/A	PostGIS polygon defining the geofence boundary
supervisor_id	UUID	36 chars	Foreign Key referencing the assigned Supervisor
description	TEXT	N/A	An optional description of the work area
created_at	TIMESTAMP	8 bytes	The creation timestamp
updated_at	TIMESTAMP	8 bytes	The last update timestamp

WORK_AREA_ASSIGNMENTS			
Field	**Data Type**	**Length**	**Description**
assignment_id	UUID	36 chars	Primary Key
workarea_id	UUID	36 chars	Foreign Key referencing the WORK_AREAS table
user_id	UUID	36 chars	Foreign Key referencing the assigned Field Worker
assigned_at	TIMESTAMP	8 bytes	The exact date and time of the

			assignment
is_active	BOOLEAN	1 byte	A flag indicating if the assignment is currently active

ATTENDANCE			
Field	**Data Type**	**Length**	**Description**
attendance_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the USERS table
workarea_id	UUID	36 chars	Foreign Key referencing the WORK_AREAS table
action_type	VARCHAR	20	Action type
timestamp	TIMESTAMP	8 bytes	The exact time of the attendance action
gps_lat	FLOAT8	8 bytes	The GPS latitude at the attendance time
gps_lng	FLOAT8	8 bytes	The GPS longitude at the attendance time
image_url	TEXT	N/A	URL to the captured selfie proof image
face_match	BOOLEAN	1 byte	Indicates if on-device facial recognition matched
is_synced	BOOLEAN	1 byte	Indicates if the offline record has synced to the server
created_at	TIMESTAMP	8 bytes	The record creation timestamp

LOCATION_TRACKING			
Field	**Data Type**	**Length**	**Description**
location_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the USERS table
attendance_id	UUID	36 chars	Foreign Key referencing the active ATTENDANCE session
gps_lat	FLOAT8	8 bytes	The GPS latitude at the tracking point
gps_lng	FLOAT8	8 bytes	The GPS longitude at the tracking point
timestamp	TIMESTAMP	8 bytes	The exact time the location coordinates were captured
is_last_seen	BOOLEAN	1 byte	Marks the last location before a geofence breach
created_at	TIMESTAMP	8 bytes	The record creation timestamp

ACCOMPLISHMENTS			
Field	**Data Type**	**Length**	**Description**
accomplishment_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the submitting Field Worker
workarea_id	UUID	36 chars	Foreign Key

			referencing the WORK_AREAS table
image_url	TEXT	N/A	URL to the work proof image
captured_date	DATE	4 bytes	The date the image was captured (from image metadata)
captured_time	TIME	8 bytes	The time the image was captured (from image metadata)
captured_lat	FLOAT8	8 bytes	GPS latitude extracted from image metadata
captured_lng	FLOAT8	8 bytes	GPS longitude extracted from image metadata
message	TEXT	N/A	An optional accomplishment message from the worker
status	VARCHAR	20	The report status (pending, approved, denied)
approved_by	UUID	36 chars	Foreign Key referencing the Admin/Supervisor who approved
approved_at	TIMESTAMP	8 bytes	The timestamp when the report was approved or denied
rejection_reason	TEXT	N/A	The reason provided if the report was denied
is_synced	BOOLEAN	1 byte	Indicates if the offline record has synced to the server
created_at	TIMESTAMP	8 bytes	The submission timestamp

LEAVE_REQUESTS			
Field	**Data Type**	**Length**	**Description**
leave_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the requesting worker
type	VARCHAR	50	The type of leave (emergency, sick, vacation, other)
start_date	DATE	4 bytes	The starting date of the leave request
end_date	DATE	4 bytes	The ending date of the leave request
reason	TEXT	N/A	The stated reason for the leave
image_url	TEXT	N/A	URL to an optional proof image (e.g., medical certificate)
status	VARCHAR	20	The request status (pending, approved, denied)
admin_decision	UUID	36 chars	Foreign Key referencing the Admin who made the decision
decision_at	TIMESTAMP	8 bytes	The timestamp of the approval/denial decision
is_synced	BOOLEAN	1 byte	Indicates if the offline record has synced to the server
created_at	TIMESTAMP	8 bytes	The submission timestamp

CREDIT_BALANCE			
Field	**Data Type**	**Length**	**Description**
balance_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the USERS table
current_score	INTEGER	4 bytes	The current performance credit score
updated_at	TIMESTAMP	8 bytes	The last update timestamp

CREDIT_HISTORY			
Field	**Data Type**	**Length**	**Description**
history_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the USERS table
change_amount	INTEGER	4 bytes	The number of points added or deducted
reason	VARCHAR	255	The reason for the change in points
created_at	TIMESTAMP	8 bytes	The transaction timestamp

NOTIFICATIONS

Field	Data Type	Length	Description
notification_id	UUID	36 chars	Primary Key
sender_id	UUID	36 chars	Foreign Key referencing the sender
receiver_id	UUID	36 chars	Foreign Key referencing the receiver
message	TEXT	N/A	The content of the notification
type	VARCHAR	50	The notification type (announcement, confidential, alert)
is_read	BOOLEAN	1 byte	A flag indicating if the notification has been read
sent_at	TIMESTAMP	8 bytes	The timestamp when the notification was sent

ALERTS			
Field	**Data Type**	**Length**	**Description**
alert_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the affected worker
type	VARCHAR	50	Alert type
message	TEXT	N/A	The alert message
created_at	TIMESTAMP	8 bytes	The alert generation timestamp
resolved_at	TIMESTAMP	8 bytes	The timestamp when the alert was resolved
resolved_by	UUID	36 chars	Foreign Key referencing the

			Admin who resolved the alert

WORKER_REPORTS			
Field	**Data Type**	**Length**	**Description**
report_id	UUID	36 chars	Primary Key
reporter_id	UUID	36 chars	Foreign Key referencing the user who submitted the report
reported_id	UUID	36 chars	Foreign Key referencing the user being reported
issue_type	VARCHAR	100	The type of issue reported
description	TEXT	N/A	A detailed description of the issue
image_url	TEXT	N/A	URL to an optional proof image
status	VARCHAR	20	The report status
created_at	TIMESTAMP	8 bytes	The report submission timestamp

SCHEDULES			
Field	**Data Type**	**Length**	**Description**
schedule_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the USERS table

start_time	TIME	8 bytes	The starting time of the shift
end_time	TIME	8 bytes	The ending time of the shift
days	VARCHAR	50	Comma-separated list of working days
created_at	TIMESTAMP	8 bytes	The schedule creation timestamp

AUDIT_LOGS			
Field	**Data Type**	**Length**	**Description**
log_id	UUID	36 chars	Primary Key
user_id	UUID	36 chars	Foreign Key referencing the user who performed the action
action	VARCHAR	20	The type of action
table_name	VARCHAR	50	The name of the affected database table
record_id	UUID	36 chars	The UUID of the affected record
old_values	JSONB	N/A	The previous values of the record
new_values	JSONB	N/A	The new values of the record (stored as JSON)
created_at	TIMESTAMP	8 bytes	The timestamp of the action

9.6 Data Integrity Rules

The database enforces several integrity rules to ensure data accuracy, consistency, and reliability. Every record has a unique ID to prevent duplicates and ensure each record can be referenced correctly. When one table references another, the database checks that the referenced record exists, preventing orphaned records. Fields are restricted to specific values—for example, attendance actions can only be time-in, break-in, break-out, or time-out, and status fields can only be active, inactive, or pending. Critical fields such as user ID, timestamps, and GPS coordinates cannot be left empty. Attendance logging is only allowed when the worker's GPS location is inside their assigned work area, and facial recognition must match the registered face before attendance is recorded. Records created offline are marked as "not synced" until successfully uploaded to the cloud, preventing data loss or duplication. All create, update, and delete actions are logged for accountability and review. Credit scores are automatically updated based on attendance and approved accomplishments, with no deductions for approved leaves, and point amounts are configurable by the admin. These rules work together to keep the database clean, reliable, and trustworthy across both online and offline operations.

9.7 Backup and Recovery Strategy

To ensure data integrity and guarantee business continuity in the event of system failure or data corruption, the system employs an automated, routine backup mechanism orchestrated through GitHub Actions. A scheduled CI/CD workflow (cron job) is configured to execute on a weekly basis, automatically triggering a secure extraction of the database schema and its records. Once the automated pipeline generates the database dump, the backup artifact is securely archived, creating a reliable, point-in-time snapshot of the system's data. By automating this process via GitHub Actions, the system eliminates the risk of human error associated with manual backups and ensures that a recent, stable version of the database is consistently available. In a disaster recovery scenario, administrators can rapidly retrieve and restore these automated backups, thereby minimizing potential downtime and preventing catastrophic data loss.

10. Testing Strategy

10.1 Testing Overview

The testing strategy defines the structured approach for verifying and validating the BantayGawaMobile before deployment. It ensures that the system satisfies its functional requirements and non-functional quality attributes while identifying and resolving defects throughout the development process.

The test shall conducted across the following levels:

- **Unit Testing:** Verifies the functionality of individual components and modules in isolation, such as QR code authentication, attendance logging, facial recognition validation, geofence validation, leave request processing, and accomplishment report submission.
- **Integration Testing:** Verifies that different modules interact correctly with one another. This includes validating the integration between QR code login and user authentication, attendance logging with facial recognition and geofence validation, accomplishment reports with proof image uploads, and report generation using attendance records.
- **System Testing:** Verifies that the complete BantayGawaMobile system satisfies all functional requirements and business rules specified in this Software Requirements Specification. This includes validating role-based access for the Admin, Field Supervisor, and Field Worker, attendance workflows, leave management, accomplishment reporting, GIS workforce monitoring, and notification services.
- **User Acceptance Testing (UAT):** Conducted with representative users, including the Admin, Field Supervisors, and Field Workers, to ensure that the system meets the operational requirements of workforce attendance monitoring and field workforce management.

Table xx. Test Case Table

Test Case ID	Test Objective	Test Steps	Expected Results	Status
TC-01	Verify that a registered user can log in using a QR code.	1. Open the application. 2. Scan a valid QR code. 3. Authenticate successfully.	User is successfully logged in and redirected to the dashboard. Session remains active for future use.	

TC-02	Verify attendance logging using facial recognition and geofence validation.	1. Log in as a Field Worker. 2. Tap Log Attendance. 3. Capture a selfie. 4. Allow GPS access.	Attendance is recorded with timestamp, GPS coordinates, and selfie. Confirmation message is displayed.	
TC-03	Verify that attendance is rejected outside the designated work area.	1. Log in as a Field Worker. 2. Move outside the assigned geofence. 3. Attempt to log attendance.	System displays "Outside geofence" error message and prevents attendance logging.	
TC-04	Verify accomplishment report submission with a proof image.	1. Log in as a Field Worker. 2. Open Submit Accomplishment Report. 3. Attach a proof image. 4. Submit.	Report is submitted successfully with status "Pending." Image contains embedded date, time, and location metadata.	
TC-05	Verify leave request submission.	1. Log in as a Field Worker. 2. Open Request Leave. 3. Fill in the required information. 4. Submit.	Leave request is submitted with status "Pending." Supervisor and Admin receive notifications.	
TC-06	Verify leave request approval by the Admin.	1. Log in as Admin. 2. Open pending leave requests. 3. Approve a request.	Leave request status changes to "Approved." Worker and Supervisor receive notifications.	
TC-07	Verify workforce monitoring by the Field Supervisor.	1. Log in as Field Supervisor. 2. Open the GIS Workforce Map.	Map displays real-time worker locations and attendance status for	

			assigned areas only. Geofence boundaries are visible.	
TC-08	Verify report generation.	1. Log in as Admin. 2. Select Generate Reports. 3. Choose a date range. 4. Generate the report.	Report is generated and displayed as preview. User can export as PDF or Excel.	
TC-09	Verify notification delivery.	1. Admin sends a notification. 2. Field Worker opens the application.	Notification appears in the worker's inbox within 5 seconds. Notification is marked as read after viewing.	
TC-10	Verify role-based access control.	1. Log in as a Field Worker. 2. Attempt to access the Manage Users module.	System displays "Access Denied" or hides the module completely. Worker cannot perform admin actions.	

11. Acceptance Criteria

REQUIREMENT	ACCEPTANCE CRITERIA
User Authentication & Role-Based Access	● The system shall allow Field Workers to complete their first-time login by scanning a unique QR code on their ID card, successfully authenticating and storing the session persistently. ● The system shall enforce mandatory Multi-Factor Authentication (MFA) for Admin and Field Supervisor logins by sending a time-sensitive OTP to the registered email upon entering valid primary credentials. ● The system shall grant access to the Admin Web Dashboard within $\leq$ 3 seconds following valid OTP entry. ● The system shall terminate active sessions automatically after 30 minutes of inactivity. ● The system shall restrict access to features based on user roles (Admin, Field Supervisor, Field Worker), ensuring that a Field Worker cannot access administrative management modules.
Workforce Profile & Geofence Management	● The Admin shall be able to create, edit, and delete user accounts, assigning roles and registering a worker's face during the account creation process. ● The Admin shall be able to create work areas on an interactive map using polygon geofencing to define irregular area boundaries. ● The Admin shall be able to assign Field Workers and Field Supervisors to one or more work areas. ● The system shall display a list of all employees filtered by work area for efficient management.
Attendance Logging & Facial Recognition	● The system shall allow a Field Worker to log attendance actions

	(Time-in, Break-in, Break-out, Time-out) only when the device's GPS coordinates are inside the assigned geofence boundary. ● The system shall verify the worker's identity through on-device facial recognition (using MobileFaceNet) before logging attendance, blocking a photo or video spoof attempt. ● The system shall capture a selfie, timestamp, and GPS coordinates for each attendance action, storing the selfie with embedded date, time, and location metadata. ● The system shall record all attendance actions and provide a success/failure response within < 3 seconds.
Field Accomplishment Reporting	● A Field Worker shall be able to submit an accomplishment report for their assigned area by capturing an image, which automatically embeds date, time, and GPS coordinates as both metadata and a visual overlay. ● A Field Worker shall be able to submit an additional accomplishment report for a different work area by manually selecting it from a map. ● A Field Supervisor shall be able to view, approve, or deny pending accomplishment reports submitted by workers in their assigned areas. ● The system shall display a clear warning if the metadata extracted from the proof image does not match the assigned work area.
Leave Request Management	● A Field Worker shall be able to submit a leave request by specifying the type, start/end date, and reason, with an option to attach a proof image. ● The system shall notify both the Field Supervisor and Admin when a new leave request is submitted. ● An Admin shall be able to approve or deny pending leave requests, and the system shall require a reason for

	denial. ● A Field Supervisor shall be able to view leave requests from workers in their assigned areas, even if they cannot approve them.
Workforce Analytics & GIS Monitoring	● The Admin and Field Supervisor shall be able to generate attendance and performance reports (Daily, Weekly, Monthly) filtered by date range, work area, and worker. ● The system shall export generated reports in PDF and Excel formats, including the report title, generation date, and applied filters. ● The system shall load the interactive GIS Workforce Map within ≤ 5 seconds, displaying real-time worker locations, color-coded attendance status, and geofence overlays. ● A Field Supervisor shall be able to view the GIS Map for their assigned areas only, while the Admin can view all areas.
Notifications & Automated Alerts	● The Admin shall be able to send in-app notifications to all users, specific roles, or specific users. ● The Admin shall be able to send confidential notifications, which are visible only to the intended recipient. ● The system shall automatically send a notification to the Admin, Supervisor, and the worker when a worker has 3 or more consecutive unexcused absences. ● The system shall deliver push notifications within < 5 seconds.
Offline Capability & Synchronization	● A Field Worker shall be able to record attendance and capture accomplishment proof images without an active internet connection. ● The system shall securely store offline attendance records and reports locally on the device using SQLite. ● The system shall automatically

	detect when internet connectivity is restored and synchronize all pending offline data (attendance, reports, leave requests) within 1 minute. ● The system shall ensure that no duplicate records are created during the automatic sync process.
Usability & User Interface	● The mobile application shall launch within < 5 seconds on devices running Android 7.0 and above. ● The application shall provide clear visual confirmation for user actions (e.g., successful check-in) within 1 second. ● The user interface shall feature large touch targets (> 44dp) for mobile use and be responsive on devices with a minimum resolution of 360dp x 800dp. ● The application shall support both English and Filipino languages with a language toggle feature. ● An in-app tutorial shall be available for first-time users and should be completable in < 5 minutes.

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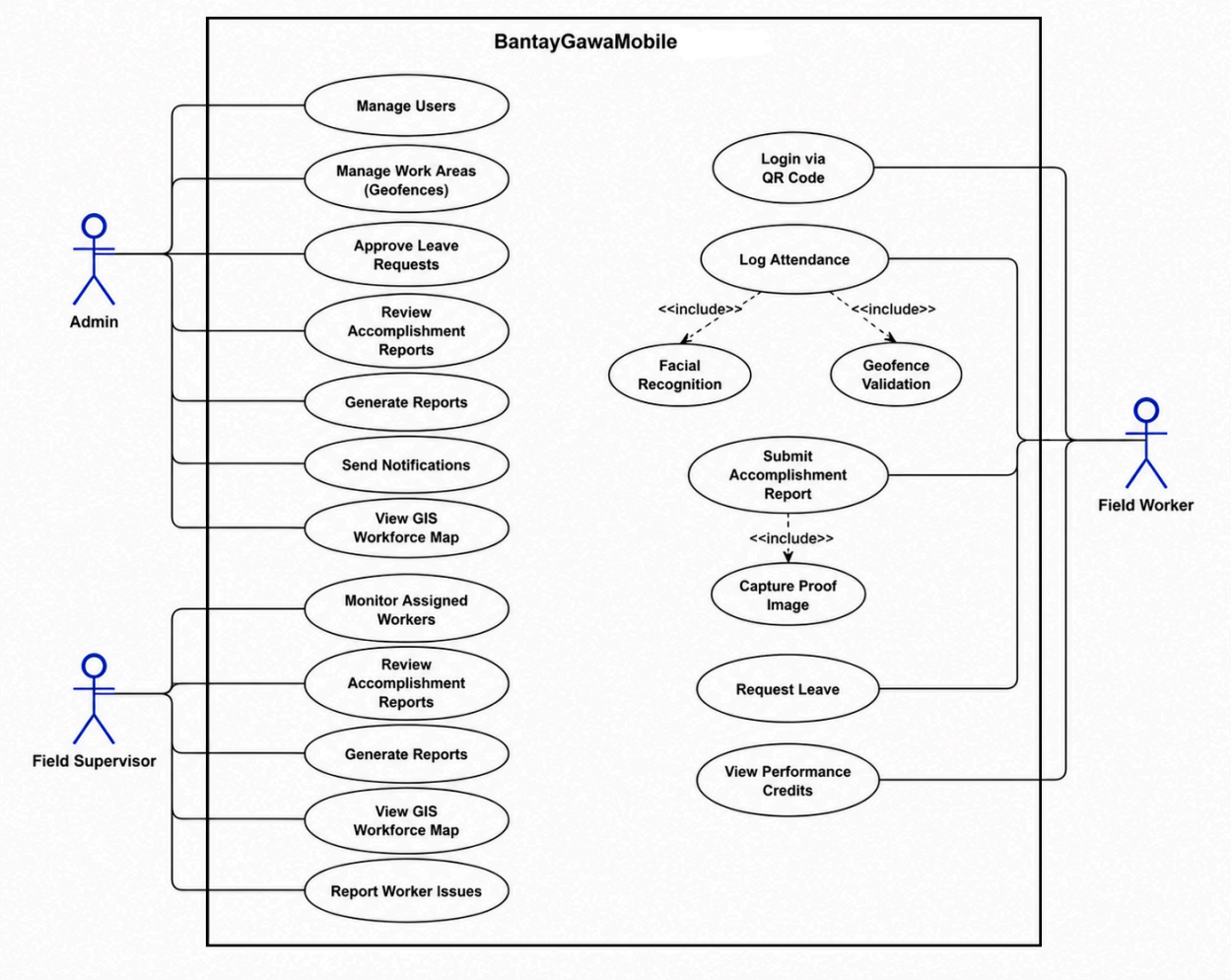
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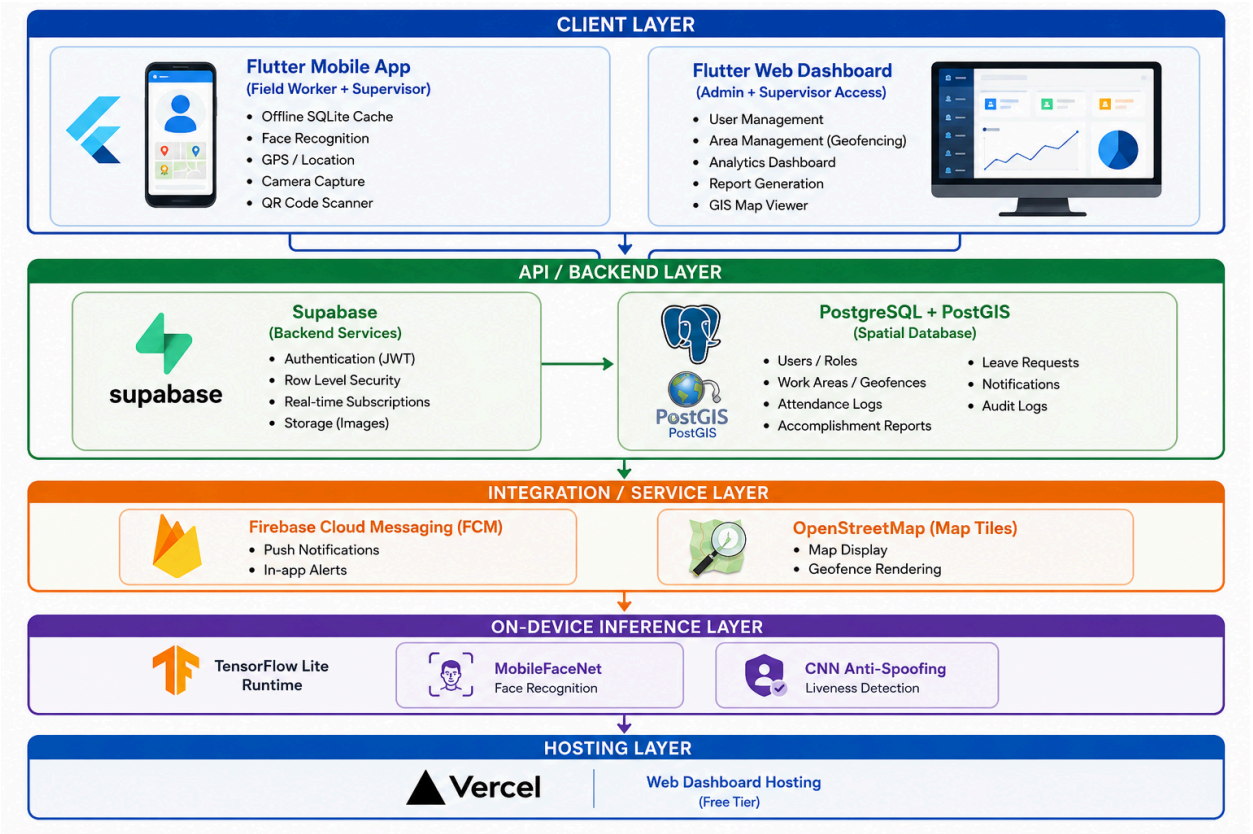
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APPENDICES

Appendices A - Use Case Diagram



Appendices B - System Architecture Diagram



Target Users

The primary users of Bantay-Gawa Mobile are the stakeholders directly involved in workforce attendance monitoring and management within the Environment and Natural Resources (ENRO), namely: Field Workers, who record their attendance through geofencing and facial recognition, submit daily accomplishment reports, and receive work assignments and notifications; Field Supervisors, who monitor field personnel, validate attendance records, review accomplishment reports, assign tasks, and provide feedback; and the System Administrator, who manages employee profiles, user accounts, work zone assignments, shift schedules, attendance records, system configurations, and report generation. Each user is assigned specific access privileges based on their responsibilities to ensure secure and efficient system operations.

Major Functions

Bantay-Gawa Mobile provides the following major functions: (1) an Employee Profile Management Module that maintains employee information, work zone assignments, shift schedules, and user accounts; (2) a Geofencing-Based Attendance Module that restricts attendance logging to authorized work locations using GPS coordinates; (3) a Real-Time Facial Recognition Module that verifies the identity of employees during attendance transactions to prevent proxy attendance; (4) a Mobile Attendance Module that records time-in, time-out, GPS location, timestamps, selfie images, and device information; (5) a Daily Accomplishment Reporting Module that enables field workers to submit completed tasks, supporting photos, remarks, and work locations; (6) a Supervisor Validation Module that allows supervisors to review attendance records, validate accomplishment reports, and provide observations or feedback; (7) a Workforce Analytics Module that generates attendance statistics, productivity reports, workforce utilization, and other analytical reports to support management decision-making; (8) a GIS Workforce Monitoring Module that displays employee locations, assigned work zones, attendance status, and completed assignments on an interactive map; (9) a Notification Module that sends attendance reminders, work assignments, schedule updates, and supervisor feedback; and (10) a User and Access Management Module that implements role-based access control for administrators, field supervisors, and field workers.

Stakeholders

This section identifies the individuals who directly interact with or are affected by the proposed Bantay-Gawa Mobile system. These stakeholders represent the intended users identified during the requirements-gathering phase and serve as the basis for defining user roles, permissions, and system functionalities.

Stakeholder	Role	Description
System Administrator	Administrator	Manages user accounts, employee records, work zone assignments, shift schedules, attendance records, reports, and overall system configuration. Ensures the security, maintenance, and proper operation of the system.
Field Supervisor	Supervisor	Monitors field personnel, validates attendance records, reviews accomplishment reports, assigns work, provides feedback, and tracks workforce performance within assigned work zones.
Field Workers	End User	Records attendance through geofencing and facial recognition, submits accomplishment reports, receives work assignments and notifications, and views personal attendance history and work schedules.

ASSUMPTIONS AND CONSTRAINTS

Assumptions

It is assumed that the field workers who will use the proposed Bantay-Gawa Mobile system have access to personal Android smartphones (version 7.0 or higher) and possess the basic digital literacy necessary to operate a mobile application. The researchers also assume that although internet connectivity may occasionally be intermittent in some work locations, the system's offline capabilities will help minimize disruptions during attendance recording. It is assumed that GPS signals are available within designated work areas to support geofencing validation and accurate location verification.

It is assumed that the Environment and Natural Resources (ENRO) has existing Human Resource (HR) or administrative personnel who will serve as system administrators,

as well as designated field supervisors responsible for monitoring field workers and validating attendance records. The researchers also assume that the ENRO can define work-area geofences using available mapping data and will provide the necessary employee information and other initial records required for system setup. It is also assumed that employees are willing to use facial recognition as part of the attendance verification process and that all users will comply with applicable privacy and data protection policies governing the collection and processing of attendance information.

Constraints

The development and implementation of Bantay-Gawa Mobile are subject to several limitations. The proposed system must be developed within the available budget of the Local Government Unit (LGU), making the use of open-source and cost-effective technologies necessary. The mobile application is intended to support Android devices running version 7.0 or higher; therefore, older devices may not fully support certain features, particularly facial recognition. Since field personnel may work in areas with unstable internet connectivity, the system is designed to allow offline recording of core attendance functions, with synchronization occurring once a stable connection becomes available. In addition, the system must comply with the LGU's data retention and privacy policies, particularly in handling attendance records, GPS locations, and biometric information. To enhance data security, facial recognition data and other sensitive biometric information will be securely stored and managed in accordance with applicable privacy and data protection regulations.

REQUIREMENT TRACIBILITY MATRIX

The Requirement Traceability Matrix (RTM) identifies the relationship between the study's objectives and the corresponding system requirements, use cases, and validation methods. It ensures that each requirement is properly addressed throughout the system's design, development, testing, and evaluation.

Objective	Requirement	Use Case	Validation Method
Manage user authentication, user accounts, and role-based access for administrators, supervisors, and field workers	FR-01, FR-02, FR-03, FR-05, FR-06, FR-07, FR-08, FR-09, FR-12	UC-01, UC-10, UC-14, UC-15	Functional testing of user authentication, account management, QR code generation, and QR login
Manage work areas and enforce geofencing for authorized attendance locations	FR-10, FR-11, FR-14, FR-15, FR-29	UC-02, UC-08, UC-10	Functional testing of work area creation, geofence configuration, and GIS visualization
Verify employee attendance using facial recognition, geofencing, and attendance logging	FR-04, FR-15, FR-16, FR-17, FR-33	UC-15, UC-19, UC-20	Functional testing of facial recognition, geofence validation, and attendance recording
Allow field workers to submit accomplishment reports with supporting evidence	FR-20, FR-22, FR-34	UC-16, UC-21	Functional and usability testing of accomplishment report submission and proof validation
Enable supervisors and administrators to review and approve submitted proofs and reports	FR-19, FR-21	UC-04, UC-10, UC-11	Functional testing of approval and validation workflows
Support employee leave request management	FR-23, FR-24	UC-03, UC-13, UC-17	Functional testing of leave request submission, review, approval, and denial
Encourage employee accountability through a workforce credit system and attendance monitoring	FR-13, FR-25, FR-36	UC-09, UC-18, UC-19	Functional testing of credit score computation, attendance monitoring, and absence alerts

Notify users of important system activities and reminders	FR-25	UC-06, UC-09	Functional testing of in-app notification delivery
Generate attendance and workforce reports for monitoring and decision-making	FR-18, FR-19, FR-26, FR-27, FR-32	UC-05, UC-12, UC-13	Functional testing of report generation and Excel/PDF export
Monitor field workers through GIS mapping and real-time location tracking	FR-28, FR-29, FR-30, FR-31	UC-08, UC-10	Functional testing of GIS monitoring and location tracking
Ensure attendance and accomplishment reporting remain available during connectivity interruptions	FR-33, FR-34, FR-35	UC-19, UC-20, UC-21, UC-22	Functional and integration testing of offline recording and automatic synchronization